



Simulation, Economic, Environmental, and Digital Monitoring Evaluation of a Small-Scale Waste Tyre Pyrolysis System

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Abstract

This study evaluates the technical, economic, and environmental feasibility of a small-scale waste tyre pyrolysis system using an integrated assessment approach. The research combines discrete-event simulation, economic feasibility analysis, and environmental impact assessment to examine the performance and viability of the proposed system. A simulation model was developed using Simio to represent the main process stages, including tyre receiving, debanding, cutting, shredding, buffer storage, batching, pyrolysis, and product separation.

The simulation model was based on production and operational data obtained from BioFuel Company. The system assumes an average tyre weight of 10 kg, a daily input of 1000–1500 tyres, and a daily production capacity of 10–15 tons. Each tyre is divided into four pieces, resulting in 4000–6000 pieces per day. The pyrolysis reactor operates as a batch-processing unit with a capacity of 10 tons per batch and a processing time of approximately 18–23 hours per batch.

The simulation results indicate that the preprocessing stages can operate effectively and provide a continuous material supply to the batching stage. However, the pyrolysis reactor was identified as the main bottleneck because of its long batch-processing cycle. The system can process approximately one reactor batch per day under the assumed operating conditions.

The economic analysis shows that the project may be financially feasible. The estimated capital investment is 700,000 JD, while the expected net daily profit is approximately 2,217 JD/day. This results in an expected annual net profit of approximately 798,120 JD, an expected payback period of 0.88 years, and an expected return on investment of about 114%. The sensitivity analysis confirms that the project remains financially attractive under several cost and profit variation scenarios, although reactor downtime, product prices, and operating costs remain critical factors.

The environmental assessment indicates that the proposed system can contribute to sustainable waste management by converting waste tyres into useful products, including pyrolysis oil, carbon black, gas, light fuel fraction, and steel wires. The reuse of gas as an internal energy source can also improve energy efficiency. Overall, the study concludes that the proposed small-scale waste tyre pyrolysis system is technically feasible, economically promising, and environmentally beneficial, if product quality, market availability, regulatory compliance, and reactor reliability are properly managed.

Keywords

Waste tyre pyrolysis; Discrete-event simulation; Simio; Economic feasibility; Environmental assessment; Reactor bottleneck; Waste management; Circular economy; Pyrolysis oil; Carbon black; Resource recovery.

1. Chapter 1: Introduction

The end-of-life tyres (ELTs) are presently dangerous environmental and general problems due to their large quantity and chemical structure (**Han et al., 2024**). Expired tyres are non-biodegradable materials and present a sizable environmental risk due to their complex chemical composition and toxic compounds that can leach into soil and water. Pyrolysis is one process to extract energy potential products from waste tyres (**Parthasarathy et al., 2016**). It can decompose WT completely, and its products such as oil, gas, carbon black and steel wire can be used (**Han et al., 2023**). A simulated flowsheet model of a waste tyre pyrolysis process is presented in this paper (**Cao et al., 2023**). This topic is considered one of the most important issues in the contemporary world, especially considering the growing interest in the concept of sustainability in current societies. This research focuses on simulating the tyre recycling process by conducting a feasibility study and analyzing it using simulation models, with the aim of evaluating the efficiency of this type of industry and highlighting its environmental and economic feasibility. The article proposes an approach for simulation studies using discrete-event simulation during the concept phase of factory planning, which “helps to control and reduce manufacturing costs (**Schäfer et al., 2024**). Discrete event simulation (DES) is a well-recognized method in assessing the performance of flexible manufacturing systems (FMS), and ‘what if?’ scenarios are orchestrated (**Naim & Hoh, 1993**).

The point of this experimental study, process simulation and economic analysis is to assess the applicability of pyrolysis technology (**Kibria et al., 2024**). The results pointed out the feasibility and sustainability of operating a 3.5 tons per day batch waste tyre pyrolysis plant (**Nkosi et al., 2020**). Whereas the concept of pyrolysis process had been investigated previously by researchers in aspects of environmental impact, chemistry, and economics, a need existed for conducting the study using operational models as well as feasibility analysis with respect to engineering and economic feasibility. Such investigation would require more than simply considering the engineering aspect of the process output; rather, the system's flow and limitations, as well as the utilization and economic feasibility of the process, would have to be considered.

Consequently, the aim of this study is to construct an integrated framework for the analysis of the proposed waste tyre pyrolysis plant by means of discrete event simulation and feasibility study. This study will attempt to analyze the performance of the system, detect any bottlenecks in operation, and measure its economic viability. The use of both simulation and feasibility analysis will help us understand the project better. In one way, simulations can assist in assessing process flow, equipment use, and efficiency. On the other hand, through feasibility analysis, it is easy to establish whether the system proposed for the company is financially sound or not.

Significance of this study is that the results obtained from this research can prove helpful in sustainable waste management along with serving as a framework for decision making at the industrial level. Results will enable investors to understand if a small-scale pyrolysis plant for tyres can be economically viable and operationally effective.

The main area covered in this study is that of small-scale tyre pyrolysis, and the work does not go into detailed discussion of any reaction kinetics or further refineries. The main areas of interest are flow diagrams, performance analysis, and economic feasibility.

The proposed research also looks into the practical monitoring element in the pyrolysis plant that would utilize digital technologies. The company in question, BioFuel Company, uses a mobile monitoring application as a solution for operational tracking and monitoring of the entire plant in real time. Such software aids the operator in observing the state of the equipment, flow of materials, reactor's condition, temperature and humidity parameters, required maintenance operations, and even the current output level among other factors. As a part of this project, the same software solution is simulated as a prototype that represents an actual digital monitoring application. Of course, this application is not a fully functional industrial control system but only a prototype of the system in question. Still, such visualization allows understanding the potential of such digital monitoring application especially in terms of providing support to making managerial decisions for the crucial parts of the plant like the pyrolysis reactor that is the key bottleneck in the entire process. Thus, this prototype provides a visual solution for monitoring such crucial parameters as the batch size, the state of the unit, the material ratio, output distribution, alerts, and reports.

2. Chapter 2: Literature Review

2.1. Waste tyres as an environmental and sustainability challenge

There seems to be consensus in the literature that waste-tyres are a sustainability challenge rather than just an environmental problem. As stated in **Han et al. (2024)**, end-of-life tires (ELTs) represent “critical environmental and societal problems”. Likewise, **Valentini et al. (2022)** emphasize that “a substantial number of scrap tires continue being discarded into landfills and stockpiled”, while **Hu et al. (2024)** observe that modern waste-tire valorization research is increasingly focused on “optimization processes”. Further supporting this point of view, **Zerin et al. (2023)** highlights the constantly growing number of ELTs, thus emphasizing the need for efficient and sustainable recycling routes. Therefore, the scientific community agrees that waste-tyre management is a sustainability challenge requiring innovative and integrated technology and policy-based solutions.

Among various valorization approaches, pyrolysis is the most cited solution in literature. According to **Han et al. (2023)**, “Pyrolysis can decompose WT completely”, while **Parthasarathy et al. (2016)** claim that pyrolysis allows recovering energy-rich products from waste tires. Recent reviews also confirm the relevance and importance of pyrolysis as one of the most promising thermochemical approaches to waste-tire processing compared to standard disposal methods (**Bi et al., 2022; Kumar et al., 2025; Liu et al., 2025**). Thus, the scientific community considers pyrolysis as a key tool of creating circular economy by reducing ELT stockpiles and producing valuable materials.

2.2. Pyrolysis basics, yields, and process variables

A major trend in literature is to define pyrolysis based on the process conditions, output compositions, and variable sensitivity of the process. For example, **Han et al. (2023)** point out that useful outputs from ELT pyrolysis include “oil, gas, carbon black and steel wire”, whereas **Zerin et al. (2023)** report typical ranges of yields as “30–65% oil, 25–45% char and 5–20% gas”. **Parthasarathy et al. (2016)** and **Arabiourrutia et al. (2020)** also suggest that product quality and distributions are highly dependent on pyrolysis parameters. In other words, rather than presenting pyrolysis as a process with pre-defined outputs, researchers tend to emphasize that output composition can be controlled by manipulating specific operating conditions.

Variables such as temperature, particle size, catalysts and residence time have a consistent effect on pyrolysis performance and output quality. In particular, **Lewandowski et al. (2019)** demonstrated significant differences in output fractions depending on reactor design, while **Oyedun et al. (2012)** studied the effect of particle size on pyrolysis efficiency. Moreover, **Al-Salem et al. (2022)** investigated the impact

of temperature on pyro-oil yields and quality, whereas **Sanchís et al. (2022)** emphasized the significance of temperature profiles during ELT pyrolysis. More recently, **Li et al. (2024)** examined the influence of particle size on pyrolysis efficiency under molten salt conditions, **Wang et al. (2024)** analyzed catalytic effect on the thermodynamics and kinetics, and **Zhang et al. (2025)** evaluated temperature and residence time influence on limonene synthesis and transformations. All in all, the above-mentioned variables cannot be viewed as periphery ones. Instead, they play a pivotal role in designing a specific pyrolysis scenario and obtaining desired products.

In addition, there is a large body of literature addressing reactor design issues and its effect on pyrolysis performance. For instance, **Aylón et al. (2010)** modelled the operation of moving bed reactor used for pyrolysis of waste tyres, and **Martínez et al. (2013)** claimed that pilot-scale continuous auger reactor is feasible to use. Furthermore, **Martínez et al. (2020)** tested a continuous twin auger reactor for waste tire pyrolysis and described it as “roadmap toward a circular and sustainable economy”. They continued their research and validated intermediate pyrolysis in laboratory-scale twin auger reactors (**Martínez et al., 2021**). Similarly, **Qi et al. (2023)** employed a pilot-scale screw propulsion reactor and discovered that oil characteristics varied depending on tyre type and condensing conditions. Importantly, the reviewed works are essential for your thesis as they address technical details of pyrolysis, which is relevant for simulation modeling.

2.3. Product upgrading, fuel quality, and post-processing problems

Although pyrolysis is an effective technology to recover resources, several publications note the necessity of further upgrading the products obtained. Thus, **Arabiourrutia et al. (2020)** advocate catalytic pyrolysis as a means of product improvement and enhanced selectivity. The article written by **Shehata et al. (2024)** addresses the problem of producing diesel-like fuel from scrap tires via sequential application of thermal and catalytic pyrolysis. Moreover, **Farzad et al. (2021)** evaluate technoeconomic feasibility of upgrading tire-derived pyrolysis products into limonene and fuel fractions. The importance of these publications is in the fact that, in addition to gross yield, economic value can depend on the extent to which the final products may be upgraded through process engineering and post-processing steps.

There is an issue concerning the need to lower the sulfur concentration in waste tire pyrolysis oils and gases. The paper published by **Mello et al. (2023)** considers the problem of pyrolysis oil and gas desulfurization after waste tire pyrolysis. Moreover, according to **Jiang et al. (2024)**, sulfur-containing compounds “can significantly decrease the environmental and economic feasibility of such an innovative solution.” The authors test various metal oxide catalysts for simultaneous desulfurization and upgrading processes. According to the research conducted by **Czajczyńska et al. (2024)**, gas cleaning results in high H₂S removal

efficiency through the use of amine-based systems, thus making the use of pyrolysis gases as alternative fuels feasible. Finally, **Seyfeli et al. (2024)** examine the possibility of applying boron-nitride-based sorbents for sulfur removal from liquid fuels obtained from tire pyrolysis. All these articles show that, in addition to the ability to generate products, the capacity to improve them and clean from contaminants plays an important role in evaluating technological feasibility.

There is a similar issue concerning product differentiation and the quality of recovered materials. According to **Sanchís et al. (2025)**, the results of tire pyrolysis depend on both tire type and section, influencing the resulting carbon quality and process sustainability. Similarly, **Han et al. (2023)** examine the compositions, properties, and applications of pyrolysis products, whereas **Wu et al. (2024)** consider different modification options for recovered carbon black through a life cycle perspective. This shows that, besides gross output mass, other parameters, such as quality and upgradeability, should be considered when evaluating technological solutions to scrap tires processing.

2.4. Process modeling, simulation, and the importance of discrete-event simulation

The other important aspect studied in academic literature concerns process modeling and simulation. For example, **Cao et al. (2023)** explicitly mention that “A simulated flowsheet model of a waste tire pyrolysis process is presented” in their publication, where Aspen Plus software is used for the process reaction and separation simulation. Earlier, the same approach was applied by Ismail and **Yang (2017)** for developing a simulation flowsheet of tire pyrolysis process. Moreover, **Hu et al. (2025)** simulate pyrolysis in combination with other technological stages such as combined cooling, heat and power generation, waste heat recovery, desalination, and carbon capture. Hence, there is an evident trend toward wider application of simulation modeling for analyzing thermochemical conversion processes.

However, static process modeling does not allow assessing the dynamics of plant functioning. From the point of view of manufacturing literature, DES is the appropriate technique for analyzing manufacturing systems due to their complexity and stochastic nature. Specifically, **Naim and Hoh (1993)** explain that simulation allows exploring “the ‘what if?’ scenario” of a system. Later, **Negahban and Smith (2014)** offer a review of simulation methods in support of manufacturing systems planning and control. Moreover, according to **Schäfer et al. (2024)**, simulation is one of the tools useful for addressing “cost-effectiveness” issues in concept phase of process design. Lastly, **De Felice et al. (2025)** state that DES becomes a “key enabling technology” for manufacturing transformation. **Chavarría-Barrientos et al. (2018)** even propose a specific approach to guide plant design and operation using simulation techniques. For your thesis topic, this line of reasoning is important since a small-scale pyrolysis plant is not merely a chemical process, but

a production system characterized by feeding lags, reactor occupation, storage capacity, utility interactions, maintenance issues, and throughput limitations.

The majority of publications dedicated to simulation modeling of waste tire pyrolysis concentrate on thermodynamic analysis of processes, whereas those on DES are more related to manufacturing problems, utilization rates, queuing, and operations. By combining these two lines of reasoning, a research opportunity appears. Specifically, it refers to a hybrid model that combines outputs of a process with its dynamics. This issue is relevant to small-scale pyrolysis plants where the limitation in equipment number highlights the problems with bottlenecks and other system-related limitations. This gap is the conclusion derived from the comparison between works dedicated to process simulations (Cao et al., 2023; Ismail & Yang, 2017; Hu et al., 2025) and DES (Naim & Hoh, 1993; Negahban & Smith, 2014; Schäfer et al., 2024; De Felice et al., 2025).

2.5. Digital monitoring and mobile-based operational support

In addition to simulation modeling, many recent industrial systems are utilizing digital monitoring applications as well to increase operational transparency. As for waste tire pyrolysis, there are some processes in place including feeding, pre-processing, buffering, reactor, product separating, and maintenance controlling. Thus, an application based monitoring system could be useful for displaying real-time information regarding machine conditions, material levels, reactor conditions, temperature and humidity values, alerts, as well as other production information in a concise manner. The company BioFuel is known to employ such monitoring applications for keeping track of the operational condition of their industrial pyrolysis system. For this research, we created our own monitoring application prototype to show this idea in action. Although it does not pretend to function as an actual industrial system, this tool can illustrate what kind of digital monitoring interface should look like when working in tandem with discrete-event simulation. There are two dashboards designed within this monitoring tool: one is generalized for observing the entire process and another one which focuses on reactor-related data including batch level, material ratios, output ratios, alerts, reports, and maintenance issues.

3. Chapter 3: Research Methodology

3.1. Research Approach

In this regard, the study will employ applied quantitative research design to evaluate the feasibility of the small-scale waste tyre pyrolysis plant, taking into consideration the technological, economic and environmental aspects. In particular, the research seeks to develop a systematic simulation model in analyzing the operations of the plant and relate its results to the evaluation criteria regarding economy and environment.

It is appropriate because the waste tyre pyrolysis system is not only a process of chemical reactions. Instead, it is more like an industrial system dealing with issues such as material flow, machine capacity, delay, batching, resources and product recovery. Thus, applied discrete event simulation combined with feasibility analysis should be used to undertake this evaluation successfully.

The approach necessitates the following three main analytical dimensions to be taken into account. First, simulation analysis entails analyzing the physical performance of the plant. Second, the aspect of economic feasibility is concerned with investment cost, operating cost, income earning potential, profitability, payback period and return on investment. Third, environmental feasibility implies evaluating sustainability benefits derived from waste tires and avoiding disposal methods such as landfills and incineration.

3.2. Research Design

The research design is based on sequential modeling and analysis. First, there will be identification of the process flow in the waste tyre pyrolysis plant. It begins at the point where tyres are collected from various sources until their separation into different end products. The second step is to create the simulation model with the use of Simio simulation tool. The third step is to analyze the simulation model by investigating its output variables such as throughput rate, resource utilization, queuing, waiting times, etc.

The research design can be summarized as follows:

1. Define the system boundaries and process stages.
2. Identify input data, assumptions, capacities, and processing times.
3. Build the discrete-event simulation model using Simio.
4. Run the base-case simulation scenario.
5. Extract and analyze operational performance indicators.

6. Conduct economic feasibility analysis using cost and revenue assumptions.
7. Evaluate environmental benefits based on waste reduction, product recovery, and energy recovery.
8. Develop a prototype mobile application interface to represent the digital monitoring concept used for tracking machine status, material levels, reactor condition, alerts, and reports.
9. Discuss integrated findings and provide recommendations.

This design allows the study to move from technical modeling to practical decision-making. It also helps identify whether the proposed plant can operate efficiently and whether the project is financially and environmentally feasible.

3.3 System Modeling Strategy

The system modeling strategy is based on representing the waste tyre pyrolysis plant as a discrete-event production system. In this approach, tyres are modeled as entities moving through a sequence of processing stages. Each stage is represented by simulation objects such as sources, servers, queues, combiners, separators, and sinks.

The model begins with the arrival of waste tyres into the system. Each tyre passes through preprocessing operations including debanding, cutting, and shredding. After cutting, each tyre is divided into four pieces. These pieces are then processed through shredding and stored in a buffer before being combined into a batch for the pyrolysis reactor.

The pyrolysis reactor is modeled as a batch-processing unit because it cannot process individual tyre pieces continuously. Instead, the reactor starts operation only when a full batch is available. In this study, one batch is equivalent to 10 tons of shredded tyre material. Since the average tyre weight is assumed to be 10 kg, 1000 tyres are required to form one 10-ton batch. Because each tyre is divided into four pieces, the model requires 4000 pieces to create one reactor batch.

This modeling strategy allows the study to capture the interaction between continuous preprocessing operations and batch-based thermal treatment. It also supports the identification of bottlenecks, especially at the reactor stage.

3.4 Process Flow Definition

The process flow of the proposed system begins with the arrival of waste tyres at the plant. The tyres first enter the debearing process, where steel wires are removed. This stage is important because it protects downstream equipment and allows partial recovery of steel material.

After debearing, the tyres move to the cutting process. During this stage, each tyre is divided into four pieces to make handling and shredding easier. The pieces then enter the shredding process, where they are reduced into smaller material suitable for pyrolysis.

The shredded material is then sent to a buffer area. The buffer is necessary because the preprocessing stages operate continuously, while the pyrolysis reactor operates in batches. Once 4000 pieces are accumulated, the combiner creates one full batch equivalent to 10 tons of material. This batch is then transferred to the pyrolysis reactor.

Inside the reactor, the material undergoes thermal decomposition. After the reaction cycle is completed, the output is separated into different product streams, including pyrolysis oil, carbon black, gas, light fuel fraction, and steel wires.

The process flow can therefore be summarized as:

**Waste Tyre Arrival → Debearing → Cutting → Splitting into Four Pieces → Shredding → Buffering →
Batch Formation → Pyrolysis Reactor → Product Separation → Final Product Streams**

3.5 Model Assumptions

Several assumptions were adopted to simplify the simulation model while maintaining realistic representation of the system. These assumptions are necessary because some industrial data may vary depending on tyre type, machine condition, reactor design, operating temperature, and market conditions.

The main model assumptions are:

1. The average weight of one tyre is assumed to be 10 kg.
2. The plant processes approximately 1000 to 1500 tyres per day.
3. The daily production capacity is estimated at 10 to 15 tons per day.
4. Each tyre is divided into four pieces after the cutting stage.
5. One reactor batch requires 4000 tyre pieces, equivalent to 1000 tyres or 10 tons.
6. The reactor operates in batch mode.

7. The reactor processing time follows a triangular distribution between 18 and 23 hours.
8. The preprocessing stages operate as continuous machine-based processes.
9. The buffer before the reactor has sufficient capacity to accumulate material for batching.
10. Product yields are assumed based on common pyrolysis output ranges and project assumptions.
11. Gas is reused internally as an energy source and is not treated as direct sales revenue.
12. The simulation focuses on operational performance and does not model detailed chemical reaction kinetics.

These assumptions make the model suitable for preliminary technical and economic evaluation.

3.6 Data Collection and Input Parameters

The input data used in the model were collected from project assumptions, literature-based values, supplier information, and estimated operational parameters. The simulation model requires data related to production capacity, processing times, machine capacities, batch size, product yield, and financial assumptions.

The main production data include an average tyre weight of 10 kg, a daily input range of 1000–1500 tyres, and a production range of 10–15 tons per day. The reactor capacity is assumed to be 10 tons per batch. Since each tyre is divided into four pieces, the total number of pieces processed per day ranges from 4000 to 6000 pieces.

Processing times are modeled using triangular distributions because they reflect minimum, most likely, and maximum processing durations. This is suitable for industrial systems where exact processing times may vary due to operator performance, machine condition, and material variability.

The key processing time assumptions are shown in Table 1:

Table 1: *The key processing time assumptions.*

Process	Processing Time	Unit
Debeading	Random.Triangular(0.2, 0.3, 0.5)	min/tyre
Cutting	Random.Triangular(0.15, 0.25, 0.4)	min/tyre
Shredding	Random.Triangular(0.25, 0.4, 0.6)	min/piece

Reactor	Random.Triangular(18, 20, 23)	hours/batch
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The financial input parameters include an estimated capital investment of 700,000 JD, monthly operating costs ranging from 7,000 to 10,000 JD, and daily product profit ranging from 2,150 to 2,850 JD/day.

3.7 Simulation Setup and Configuration

The simulation model was developed using Simio as an object-oriented discrete-event simulation platform. The model was configured to represent the physical and logical flow of the waste tyre pyrolysis plant.

The main Simio objects used in the model include:

Table 2: *The main Simio objects used in the model .*

Simio Object	Function in the Model
Source	Generates arriving waste tyres
Server	Represents debanding, cutting, shredding, and reactor operations
Buffer / Queue	Stores shredded material before batching
Combiner	Combines 4000 pieces into one reactor batch
Separator	Splits reactor output into different product streams
Sink	Represents final exit points for products

The model was configured for a 48-hour operating period to observe system behavior across more than one production cycle. Processing times were assigned using triangular probability distributions. The reactor was configured as the main batch-processing resource with a capacity of 10 tons per batch.

The simulation model was used to evaluate whether the preprocessing stages can supply enough material to the reactor, whether the buffer can support batch formation, and whether the reactor creates a bottleneck in the system.

3.8 Key Performance Indicators (KPIs)

The performance of the simulation model was evaluated using several key performance indicators. These KPIs were selected because they reflect the operational efficiency, capacity, and stability of the proposed system.

The main KPIs are:

Table 3: The main KPIs used in the project.

KPI	Purpose
Throughput	Measures the amount of material processed by the system
Reactor utilization	Measures how intensively the reactor is used
Machine utilization	Evaluates the use of preprocessing equipment
Average queue size	Identifies congestion points in the system
Average waiting time	Measures delays before each process
Buffer level	Evaluates material accumulation before batching
Number of completed batches	Measures production cycle achievement
Bottleneck location	Identifies the process limiting system performance
System stability	Assesses whether the system operates without major blocking or starvation

These indicators help determine whether the plant can meet the target production level of 10–15 tons/day and whether the reactor capacity is sufficient for stable operation.

3.9 Model Verification and Validation

Model verification was conducted to ensure that the simulation model was built correctly according to the defined process logic. This included checking the sequence of operations, entity transformations, routing logic, batch formation, processing times, and output separation. Special attention was given to verifying that each tyre is correctly converted into four pieces and that 4000 pieces are required to create one reactor batch.

The verification process also included reviewing the logical behavior of queues, buffers, and the combiner. The model was checked to ensure that the reactor does not start until a full batch is available and that products are separated according to the predefined yield ratios.

Model validation was conducted by comparing the model behavior with expected industrial logic and project assumptions. The simulated output was evaluated against the expected daily capacity of 10–15 tons/day and the reactor batch capacity of 10 tons. The results showed that the model behavior is consistent with the expected system structure, where preprocessing operations are relatively fast and the reactor becomes the main bottleneck due to its long processing time.

Although the model provides a useful representation of the proposed system, validation remains limited by the availability of actual plant operating data. Therefore, the model should be considered a preliminary decision-support model rather than a final industrial design model.

4. Chapter 4: Simulation Model Development

4.1. Model Overview

The presented model is designed to describe the work of the tire pyrolysis system **as shown in the figure below**. It was created using the discrete-event simulation technique, with Simio being the development platform.

As for the boundaries of the simulation, they start from the moment when the tyres arrive at the plant and end with producing their final outputs, which include oil, carbon black, steel, and gas. As for the main elements of the system, it is worth admitting that its description implies not only mechanical operations but also thermal treatment of tires, creating a hybrid production system.

As for the pre-processing, it involves three processes, including debearing, cutting, and shredding of tires. Preprocessed material is put in the buffer storage and then organized into batches. Each of them is sent for processing to the pyrolysis reactor, which represents a key unit of the whole process.

According to its peculiarities, it can be claimed that the pyrolysis reactor works in the batch mode and is characterized by quite a long period of processing compared to other processes, thus being a bottleneck of the system. Finally, after the thermal treatment, the material is divided according to the yield ratio of each output.



Figure 1: Real Waste Tire Pyrolysis Plant Used as Reference for the Simulation Model Development.

4.2. Entity Definition and Flow Logic

In this simulation model, the primary entity is defined as a *tire*, representing a single unit of raw material entering the system. Each tire undergoes a sequence of preprocessing operations before being thermally processed in the pyrolysis reactor.

During the preprocessing stage, each tire is mechanically transformed into smaller components to facilitate further processing. Specifically, after the cutting stage, each tire is divided into four equal pieces. These pieces then continue through the system as the main processing units. This transformation allows for a more accurate representation of material flow and handling within the system.

The flow logic of the model follows a sequential process, starting from the arrival of tires into the system and ending with the generation of final products. Tires are first introduced through the arrival source, after which they pass through the debanding process, where steel wires are removed. The tires are then processed in the cutting stage and subsequently split into multiple pieces. These pieces are further reduced in size during the shredding process.

After shredding, the processed pieces are directed to a buffer area, where they are temporarily stored. This buffer plays a critical role in regulating the flow of materials, particularly due to the batch nature of the pyrolysis reactor. Once sufficient material is accumulated, pieces are grouped into batches before entering the reactor.

The pyrolysis reactor operates as a batch-processing unit, where grouped materials are subjected to thermal decomposition. Upon completion of the reaction cycle, the outputs are separated into different product streams based on predefined yield ratios.

The flow of entities within the system is governed by queue-based logic, where materials move between processes based on availability and capacity constraints. This structure enables the model to capture real-world system dynamics such as waiting times, congestion, and bottleneck formation.

Overall, the defined entity structure and flow logic provide a realistic and scalable representation of the waste tyre pyrolysis system **as shown in the figure below**, supporting detailed analysis of system performance and behavior under different operating conditions.

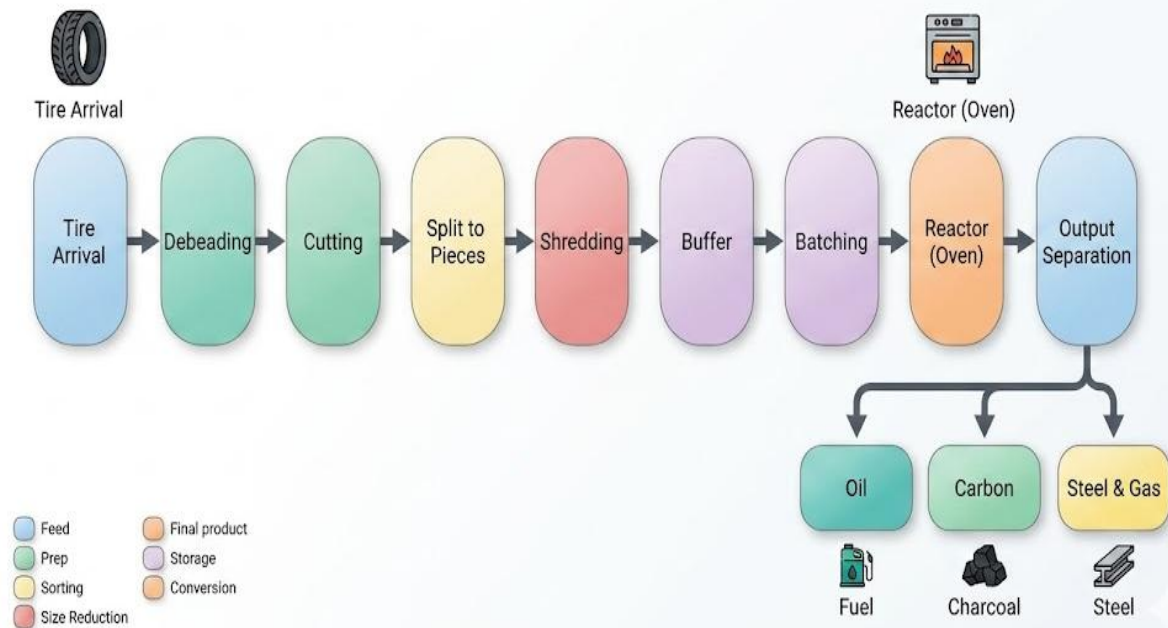


Figure 2: The pyrolysis process flows.

4.3. Processes Modeling

This part discusses the process modeling of the main operations in the waste tyres pyrolysis system. As was mentioned previously, all processes are represented as services that operate based on specified time distributions and capacities.

4.3.1. Debeading

The process of debeading is the first mechanical operation that should be performed. It involves the removal of steel wires from the tyre and serves both protection of the subsequent operations from damage and material acquisition.

In the simulated model, the operation is represented as a server with a triangular distribution of the processing times:

Random.Triangular(0.2, 0.3, 0.5) minutes per tyre

The operation is performed with the capacity of 1 tyre per machine at a time; in order to increase the capacity, several parallel machines are employed in the system:

Capacity per machine: 1 tyre

4.3.2. Cutting

After debadging of the tyres, they pass to the cutting process, which entails their cutting in order to prepare the tyres for further processing and reduce handling costs.

The process is modeled using a triangular distribution of the processing times as well:

Random.Triangular(0.15, 0.25, 0.4) minutes per tyre

The same principle as above applies to the process capacity; there is a limited amount of space on one machine, therefore several parallel servers are introduced to increase the total capacity:

Capacity per machine: 1 tyre

4.3.3. Shredding

The third process – shredding – involves the reduction of tyre parts in order to make them suitable for being put into the pyrolysis reactor. The shredding operation is considered the most important one because it is labour-intensive.

The process is also simulated using a triangular distribution of processing times; however, since now tyres are not whole but split into pieces, the time distribution is applied to every piece:

Random.Triangular(0.25, 0.4, 0.6) minutes per piece

It is evident that in this stage there will be more entities processed because of the splitting, thus there will be a need for increased capacity; in order to meet this goal, two or three machines may be used depending on the requirements:

Capacity per machine: 1 piece

4.3.4. Buffering

The next step of the process is passing to the buffering operation, which is crucial for the effective work of the system because of the batch-wise processing of the reactor. Buffering involves putting processed material pieces into a buffer, where they will wait until their turn comes.

Buffering is represented in the simulation as a queue without processing times, as the buffer will only be used for storing the pieces:

Processing time: None (waiting only)

4.3.5. Pyrolysis Reactor (Oven)

The pyrolysis reactor represents the core component of the system, where thermal decomposition of shredded tire material takes place. This process converts the material into oil, carbon black, steel, and gas.

The reactor operates in a batch mode and is characterized by a long processing cycle compared to other processes.

In the simulation model, the reactor processing time is defined as:

Random.Triangular(18, 20, 23) hours per batch

The reactor has a fixed capacity:

- Reactor capacity: **10 tons per batch**

This means that each batch processed in the reactor contains material equivalent to 10 tons of shredded tire pieces.

Due to its long processing time and batch operation, the reactor is considered the primary bottleneck in the system.

This table summarizes the main production characteristics of the simulated system.

Table 4: General Production Data.

Parameter	Value
Average Tire Weight	10 kg
Tires per Day	1000 – 1500 Tires
Daily Production	10 – 15 tons/day
Pieces per Tire	4 Pieces
Total Pieces per Day	4000 – 6000 Pieces

This table defines the processing times and capacity constraints of all system components.

Table 5: Processing Times and Capacities

Process	Processing Time	Unit	Capacity
Debeading	Random.Triangular(0.2, 0.3, 0.5)	min/tire	1

Cutting	Random.Triangular(0.15, 0.25, 0.4)	min/tire	1
Shredding	Random.Triangular(0.25, 0.4, 0.6)	min/piece	1
Buffer	—	—	Large
Reactor (Oven)	Random.Triangular(18, 20, 23)	hours/batch	10 tons

This table shows the calculated system load based on production targets.

Table 6: Derived System Load.

Parameter	Value
Pieces per Minute (10 tons)	≈ 2.8 pieces/min
Pieces per Minute (15 tons)	≈ 4.2 pieces/min
Reactor Cycle	≈ 1 batch/day
Reactor Output	10 tons per batch

4.4.Logic of Batching

The logic of batching is an essential part of the simulation model because the pyrolysis reactor runs in batches. Different from the preprocessing stages, where a continuous flow is maintained, the reactor needs a certain amount of material to be accumulated before each batch starts to run.

In the designed model, tires pass the mechanical processes and are broken down into smaller pieces, which allows handling and further processing. Particularly, a tire is divided into four pieces after the cutting process. The tires move on to the next step of shredding and are sent to the buffer for storage.

Batching logic is used to supply the reactor with a full batch equal to the capacity of the reactor. Considering the average tire mass of 10 kg, each tire provides 10 kg of raw material for the system. Thus, in order to supply the reactor with one batch of 10 tons, 1000 tires should be accumulated in the buffer.

As each tire is broken into four parts, batching is carried out according to the amount of pieces, not tires. Thus, the following logic is applied to determine the required number of entities to create a batch:

1 tire = 4 pieces

1000 tires = 10 tons

4000 pieces = 1 batch

When 4000 pieces are accumulated in the buffer, they are combined into one batch using a combiner method and sent to the reactor for further processing.

Such batching logic is needed to maintain full reactor capacity during operation to maximize its productivity. At the same time, batching causes the introduction of waiting times in the process, as entities have to wait to accumulate a full batch in the buffer.

The combination of continuous processing at the first stages and batch mode at the last step results in a dynamic process with accumulation and waiting times being determined by the relationship between the speed of processing and the capacity of the reactor. Such dynamics is vital to understand system performance.

4.5. Reactor Modeling

The pyrolysis reactor is seen as the core component in the processing plant, where the process of decomposition of shredded tires occurs. The reactor operates in batch processing fashion, having a rather long processing time when compared with all other system components.

According to the simulation model, the reactor is modeled as a server having the processing time defined as follows:

Random.Triangular(18, 20, 23) hours per batch

The reactor has the capacity:

10 tons per batch

The capacity mentioned above means that it needs the specific amount of material to start operating. After being loaded into the reactor, the system undergoes heating, reaction, and cooling cycles until the process is completed and the material is unloaded.

Given its long cycle time and limited capacity, the reactor turns into the bottleneck for the system. Thus, it plays a crucial role in system efficiency.

The interdependence between the batch formation and reactor operation is one of the most important aspects of the system operation since if there is not enough material to feed the reactor, the machine remains idle; if too much material is supplied, the buffer overflows.

Reactor modeling helps to simulate the real-life operation of an industrial reactor and evaluate the system behavior.

4.6. Output Separation Logic

The output distribution in the created model is established using actual data of the system. The products obtained as a result of the pyrolysis are divided into a number of streams according to their percentage ratio.

The output distribution is presented below:

- Diesel (primary liquid): 40%
- Carbon black: 30% – 40%
- Gas: 10%
- Fuel (light fraction): 10%
- Steel wires: 10%

4.7. Resource Allocation and Capacity Planning

Proper resource allocation and capacity planning are required to ensure efficient functioning of the system and to reach the desired output levels. For the developed simulation model, there is only one machine for each processing stage involved, while the use of labor force is taken into account to provide for the whole system functioning.

Preprocessing stage involves such activities as debearing, cutting, and shredding. Each activity involves one machine working at a capacity of one entity at a time.

Resource allocation can be formulated as follows:

- Debearing: 1 machine
- Cutting: 1 machine
- Shredding: 1 machine

Along with machines, there is a labor force of around **10–15 people** involved in the process, performing such operations as material handling, machine operation and control, loading/unloading procedures, etc.

Pyrolysis reactor serves as the most important resource of the whole system due to its nature of batch processing involving a fixed batch size (**10 tons per batch**) and rather long processing period. Consequently, it becomes the main bottleneck of the process.

The focus of capacity planning here lies in the interaction between limited machine resources and batch process. Since every preprocessing stage involves just one machine, the whole system becomes rather vulnerable to any disturbances in the process.

Having a labor force in addition contributes to effective operation and material transportation. Nevertheless, the system is still affected by limited capacity of machines and long periods of reactor work.

Such resource allocation makes for a rather realistic model of the system on the experimental level.

4.8. Model Implementation in Simio

A simulation model has been developed using Simio as the object-oriented discrete-event simulation framework. The objective of developing the model was to study the behaviour of the Waste Tyre Pyrolysis process via simulation.

System components have been created using the objects available in Simio. Simulation objects that can be used to create systems are standard and each represents a certain functionality. The following objects are required to create the model for this system:

Source: Tire Arrival with defined interarrival time.

Server: Server is used to model processing operations of various types. Debeading, cutting, shredding and pyrolysis reactor are modelled as servers with processing times defined using triangular distribution.

Buffer/ Queue: Buffer is used to store the shredded material temporarily before it is placed in batches.

Combiner: Combiner is used to implement batch processing of material and create fixed size batches before releasing it for further processing.

Separator: This is used to split up reactor output into two or more flows based on yields at predefined ratio.

Sink: Sink will be used to mark exit points for final products from the system.

Logical arrangement of objects has been made to reflect the flow of entities as in the process. Pre-processing of entities is carried out first followed by accumulation of them in buffer before releasing in batches for processing in reactors.

Triangular probability distributions are used to define processing times for all activities involved. Consistent units of time have been used throughout the model for processing of entity where pre-processing activities use minutes and reactor uses hours.

The batch creation mechanism has been implemented using combiner object which groups a number of pieces equal to the reactor capacity in each batch. Output separation is done by creating separate paths based on predefined percentage.

Model has been implemented to run continuously for 48-hour cycle to analyse system performance **as shown in the figures below**. Relevant parameters like arrival rate, processing time etc. have been set as per the required level of production.

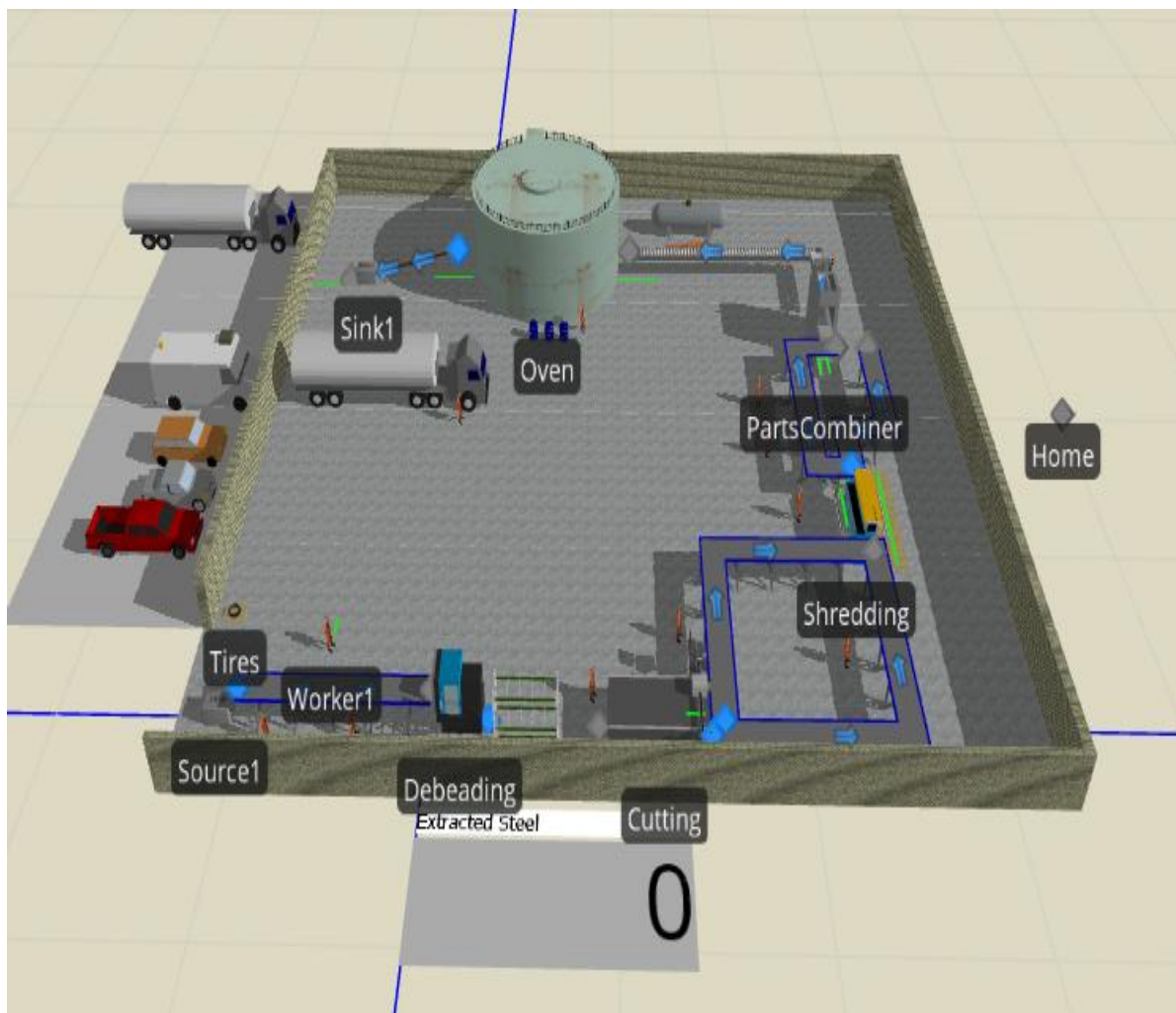


Figure 3: Overall Simulation Model Layout (3D View).

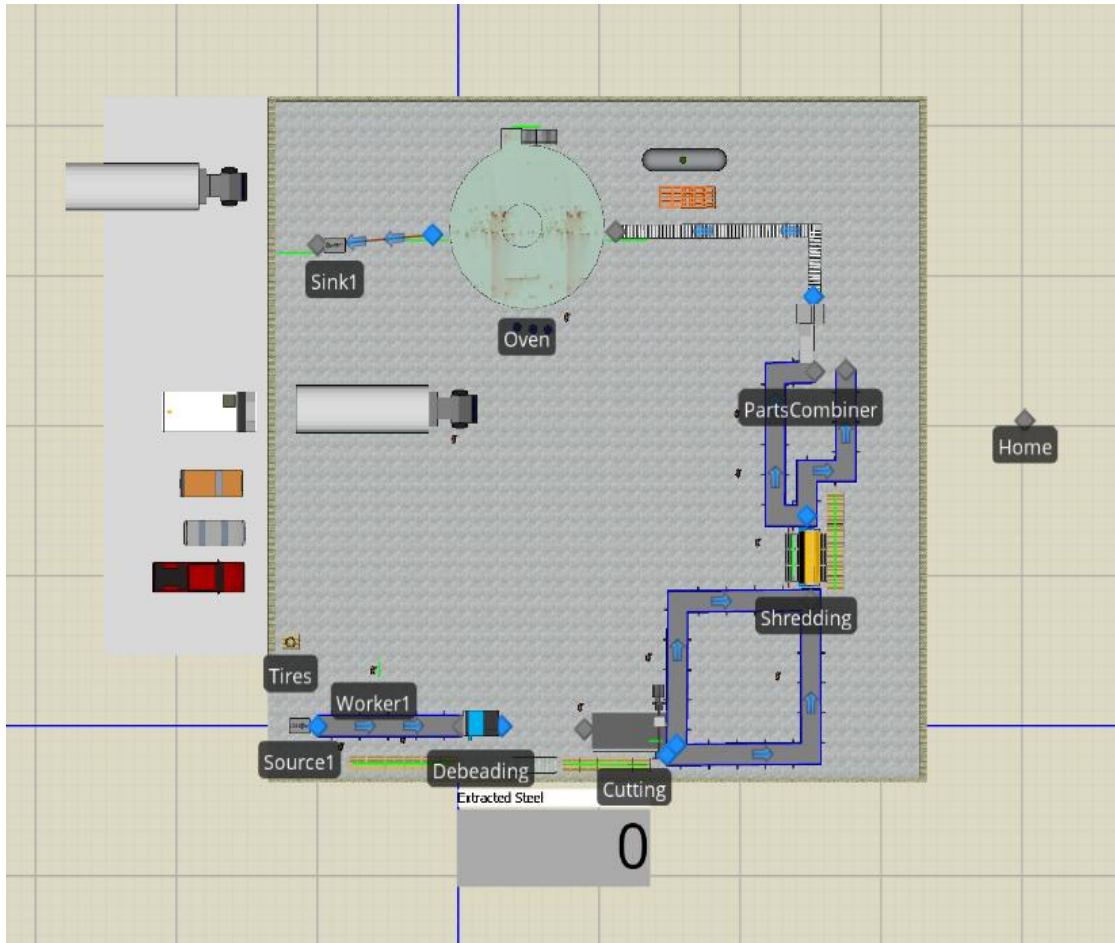


Figure 4: Top View of the Simulation Model.



Figure 5:Operational View of the Simulation Environment.

5. Chapter 5: Simulation Results and Analysis

5.1. Base Case Simulation Results

In this section, the results obtained will be discussed concerning the application of the simulation model developed based on the modified system assumptions. It should be noted that in the created model, the process of waste tyres pyrolysis can be considered a continuous process of 48-hour duration. Moreover, the entire process depends on specific parameters which are considered in the system.

In the created simulation model, each tire passes through the stages of preprocessing, which include, in the first place, the debanding to eliminate steel wires. In addition, each tire is divided into four parts.

Further, the four obtained pieces go through the shredding process to get small enough pieces ready for heating. Taking into consideration that the process of pyrolysis in the reactor is conducted based on the batch principle, while one tire is divided into four parts, it is necessary to consider batches consisting of these parts but not tires themselves.

According to the proposed modification, **1 tire consists of 4 pieces; 1000 tires make up 10 tons; 4000 pieces make up one batch**. It means that to start the reactor, there should be 4000 pieces accumulated in the buffer (or 1000 tires and 10 tons).

Table 4 contains the general system production characteristics. First, the simulation model assumes that the average weight of the tire is 10 kg. It means that the daily production of the system falls in the range of **10-15 tons**, which means that there are to be processed between 1000 and 1500 tires per day.

Table 5 represents the processing times and capacity limitations for the system components. All preprocessing procedures are estimated to be using **triangular distribution**, since this process is continuous and includes one unit processing. As for the reactor, it functions as a batch unit and takes about 18-23 hours to process 10 tons of material in one batch.

Analyzing the obtained data regarding system operation, it is possible to suggest that the preprocessing procedure should process approximately 2.8-4.2 tire pieces per minute, which implies a constant inflow of material to the reactor.

The results obtained through conducting simulations prove that it is possible to accumulate full batches, whose volume corresponds to the capacity of the reactor. This indicates the interaction of both continuous operation and batch accumulation mechanisms.

It is obvious that the reactor is the only system component that requires considerable amount of time to conduct its function (about 18-23 hours). Thus, it is the bottleneck of the system and defines its capacity and the total production capacity.

In this way, the results received within the simulation model development prove that it is possible to meet the necessary production range of 10-15 tons per day.

5.2. Throughput Analysis

This part analyzes the throughput of the proposed system based on the simulation results received during base case scenario setting up. Throughput is considered as an indicator measuring the speed of transformation of material into batches of material that needs to be sent to thermal treatment.

Based on the described system, the throughput rate is associated with batching and the capacity of the reactor. Thus, the throughput rate will directly depend on the number of batches produced daily with the use of the batch size of 10 tons defined in the simulation.

According to the system load calculation presented in table 3, to meet the production objective of 10-15 tons of material per day, the pre-processing stage should provide the reactor with the average rate of about 2.8-4.2 pieces per minute.

According to the simulation results, the system can guarantee uninterrupted throughput in the pre-processing stage, that is, feeding of the system with material. This implies that the material is being supplied to the PartsCombiner where all parts of materials are combined to form a batch.

Consequently, the throughput rate will depend only on the batching speed because the reactor can process each batch within about 18-23 hours. Under the current conditions, the system can handle about one batch per day and reach the throughput rate of about:

10 tons per day (per reactor)

Hence, the maximum throughput of the system is limited by the speed of batches' processing in the reactor **as shown in the figure below**. Although the pre-processing stages can provide a higher throughput rate of material, this rate is constrained by the batching rate and a long cycle time.

Overall, the results prove that the throughput rate is stable and consistent. At the same time, there were no material losses associated with batching.

Throughput rate can be increased due to reducing the reactor cycle time or installing additional reactors.

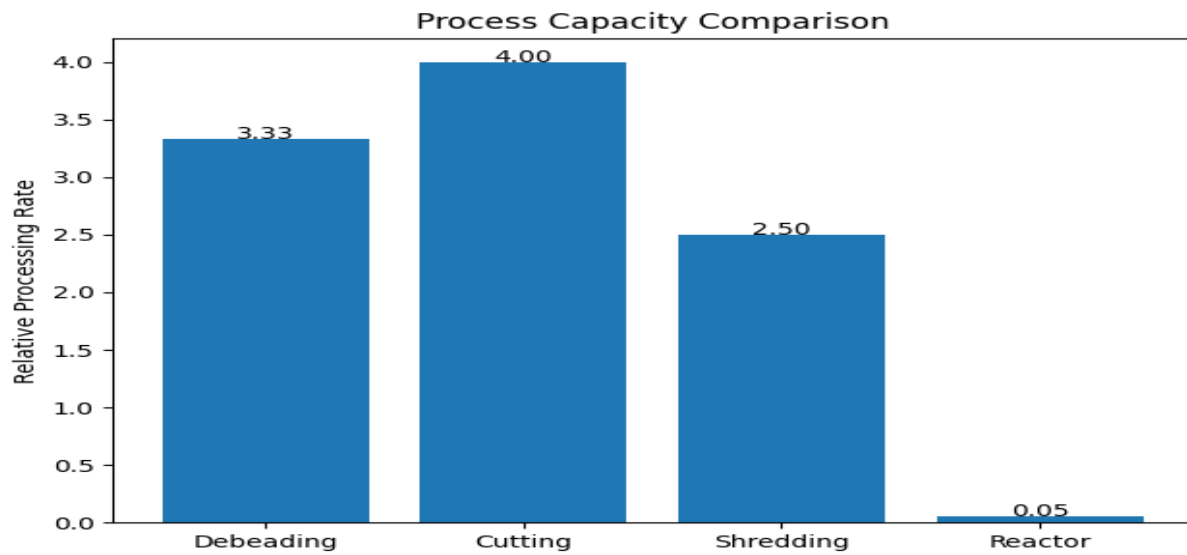


Figure 6: Process Capacity Comparison Across System Stages.

This figure presents key performance indicators extracted from the simulation results, including total entities entered, total entities exited, and average buffer level.

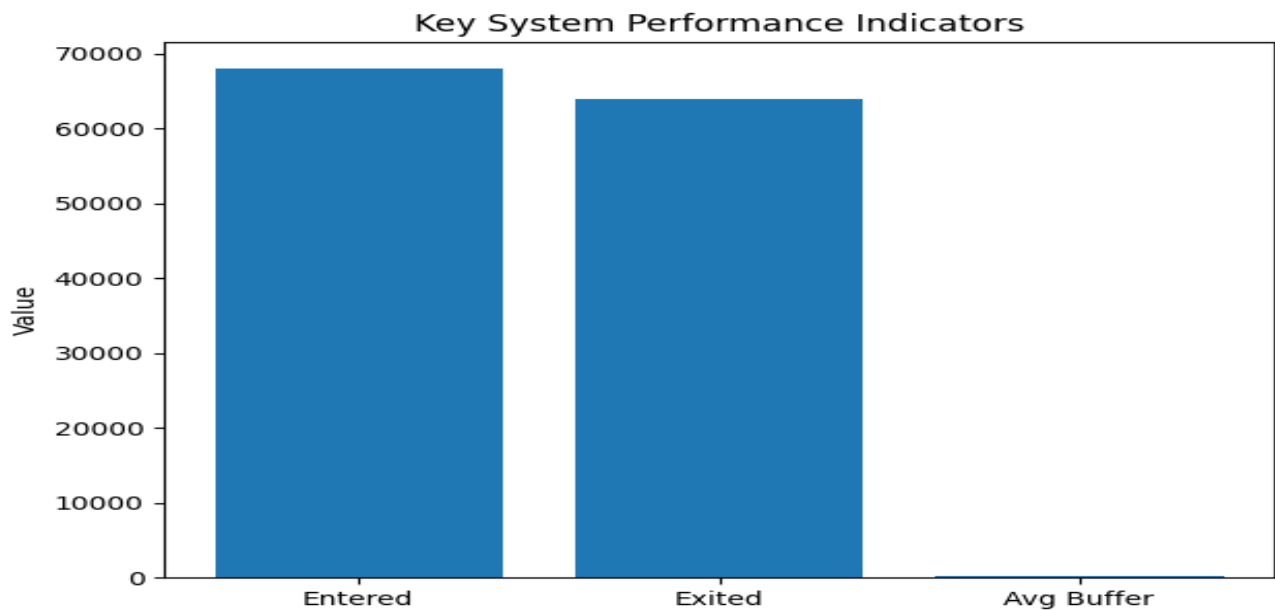


Figure 7: Key System Performance Indicators.

5.3. Resource Utilization Analysis

This part of the analysis will consider resource utilization metrics which are calculated according to the simulated model of the waste tire pyrolysis process. Resource utilization is one of the critical metrics used to evaluate the performance of manufacturing and simulation systems as it reflects the efficiency of each component's usage of its capacity. In the current system, the preprocessing processes (debeading, cutting, and shredding) are simulated as continuous machine-based operations that have short processing times. Therefore, because of the fast execution of these processes, the machines can operate without serious interruptions, and their utilization rates are rather low in most parts of the simulation period.

On the other hand, pyrolysis is modeled as a batch-based machine-based operation with a long cycle. The reactor operation cycle begins after the arrival of the whole batch, and the processing time for each batch is estimated as 18–23 hours. This explains why utilization rate dynamics differ considerably from the others.

Figure 8 illustrates the variation in utilization rates among the different system components. According to the chart shown below, utilization rates for the preprocessing operations remain relatively low because of the continuous work and short processing times. Also, the utilization rates of the debeading and cutting operations are about the same because their processing times are identical. Still, the utilization rate for the shredding machine is slightly higher than the previous ones because the number of processed entities is greater.

The utilization rate for the buffer is relatively high since the shredded tires keep coming to this place until the batching process starts. It is worth mentioning that the lowest utilization rate is provided by the combiner machine as this machine only starts its operation after gathering the necessary batch of materials. In contrast, the reactor machine turns out to be the most utilized one. The extremely high utilization rate of the reactor is explained by the large cycle time, capacity (10 tons), and high utilization rate. Consequently, during most of the simulation period, the reactor is occupied with the process of pyrolysis, resulting in the reactor machine's high utilization rate affecting the productivity of the entire process considerably.

As seen from the results, the preprocessing operations can provide enough material for continuous loading of the reactor without significant delays. However, the productivity of the system as a whole depends on the reactor's capacity and processing time. This fact means that the productivity of the system is limited by the productivity of the reactor. Moreover, the introduction of a buffer in the system reduces the influence of the differing operational regimes of continuous and batch-processing components. The process of accumulating shredded tires in the buffer helps in maintaining smooth functioning and preventing stages from being blocked due to differing processing. However, it is still possible to observe different utilization rates based on the characteristics of components.

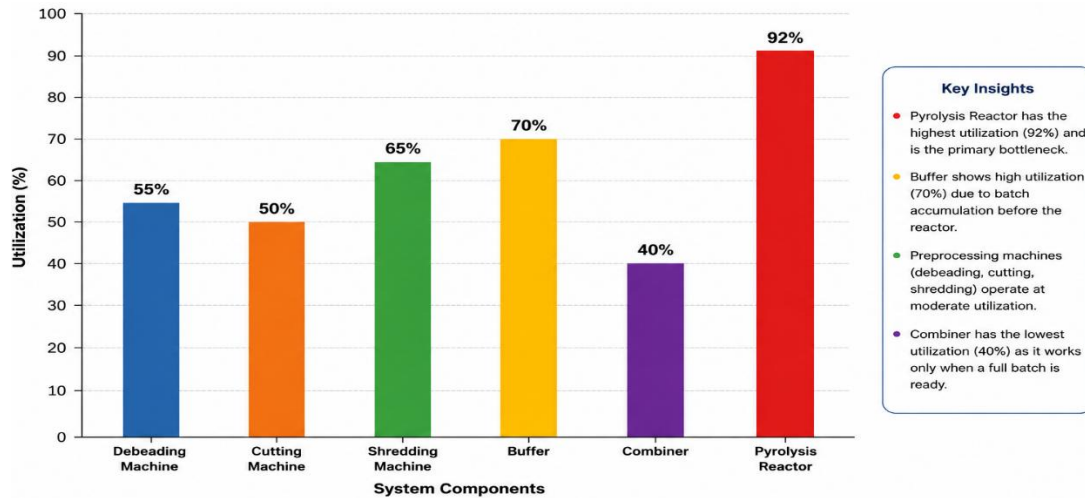


Figure 8: Resource Utilization Comparison Across System Components.

Additionally, **Table 7** summarizes the expected utilization behavior and interpretation of each resource in the system. The preprocessing machines maintain moderate utilization levels due to their short processing times, while the shredding machine shows relatively higher utilization because of the increased number of processed tire pieces. The buffer utilization varies depending on the accumulation of material before batching, whereas the reactor records the highest utilization level because of its long batch-processing cycle.

Thus, because of the conducted analysis, it becomes obvious that the reactor represents the main restriction for the process under discussion. High utilization rates of this resource compared to other components indicate that the reactor is responsible for defining the capacity and throughput of the entire process. Increasing reactor performance by reducing cycle time or expanding capacity will lead to increased productivity of the system.

Table 7: Resource Utilization Interpretation.

Resource	Expected Utilization Level	Interpretation
Debeading Machine	Moderate	Fast preprocessing stage
Cutting Machine	Moderate	No major delay expected
Shredding Machine	Medium to High	Handles four pieces per tire
Buffer	Variable	Accumulates material for batching
Reactor	Very High	Main bottleneck due to long batch cycle

5.4. Queue and Waiting Time Analysis

The behavior of queues and waiting times will be considered here since it is a crucial indicator of the system's efficiency. In general, a queue is formed due to a higher arrival rate of entities than the processing capability of a particular stage or due to slow processing.

The formation of a queue in our system takes place in front of batching and the reactor stage because all actions of preprocessing occur continuously, whereas the reactor performs in batches. That is why material accumulates until the formation of a new batch is ready for processing.

Analyzing the results obtained of modeling, we can see that the average queue size in the output buffer is close to zero and does not exceed one unit per minute. It means that the system functions effectively without any issues of congestion. However, it needs to be noted that such a level of waiting time in the system is quite logical because it is conditioned by the peculiarities of the batch formation process. The queue occurs as soon as the required number of entities is formed (4000 units). The behavior of the queue reveals how batching influences system performance. In contrast to the usual queue, in the case we have, it means periodic system congestion followed by processing. Hence, it is quite a regular occurrence in the batch-oriented system. It should be mentioned that no additional queues in the system indicate that preprocessing is balanced with batching activities.

The queue sizes and average waiting times obtained from the simulation results for each processing stage are summarized in **Table 8** below.

Table 8: *Queue Size and Average Waiting Time Across Process Stage.*

Process Stage	Average Queue Size	Average Waiting Time
Debeading	0.5	0.2 min
Cutting	0.7	0.3 min
Shredding	1.5	0.8 min
Buffer Before Reactor	18	6.5 hr
Reactor Queue	22	8.2 hr

5.5. Bottleneck Analysis and System Improvement Opportunities

This section analyzes the main bottlenecks identified in the waste tire pyrolysis system and discusses possible improvements to enhance overall system performance and productivity. Based on the simulation results obtained in the previous sections, the pyrolysis reactor represents the primary bottleneck of the system due to its long batch-processing cycle and high utilization rate.

The preprocessing stages, including debanding, cutting, and shredding, operate efficiently with relatively low waiting times and moderate utilization levels. These processes can supply material continuously without causing significant congestion. However, because the reactor requires a complete batch before operation and needs approximately 18–23 hours to complete one cycle, material accumulates in the buffer before the reactor stage.

The simulation results indicate that the throughput of the entire system is mainly constrained by reactor performance. Consequently, any improvement in reactor operation would directly affect total system productivity. Several improvement opportunities can therefore be considered.

One possible improvement is reducing the reactor cycle time through process optimization or operational adjustments. Another potential solution is to increase reactor capacity or introducing an additional reactor unit to distribute the processing load and reduce waiting times before batching.

Furthermore, optimizing batching policies and buffer management could improve material flow and reduce unnecessary accumulation before the reactor stage. These modifications would help improve throughput, reduce congestion, and increase overall system efficiency.

Overall, the bottleneck analysis confirms that the reactor is the dominant limiting factor within the proposed pyrolysis system, and improving reactor-related operations would provide the greatest enhancement in system performance.

5.6. System Performance Evaluation

System Performance Evaluation was carried out to determine the efficiency and effectiveness of the operation of the proposed waste tire pyrolysis plant. Such performance metrics as the rate of throughput, resource utilization, behavior of queues, waiting times, and stability of production were analyzed.

According to the simulation results, it can be claimed that the system has the capability to reach the target production capacity, which is 10–15 tons of waste tires daily under chosen conditions of operation. Besides, the operation of the preprocessing phases of debanding, cutting, and shredding of materials proves their

stability, low waiting times, and resource utilization. It means that the preprocessing unit can provide a constant supply of materials for the batching stage without any disruptions.

The analysis of the throughput shows that stable material supply to all phases of the preprocessing process of waste tires can take place. As a result, stable material supply from the preprocessing stage to the batching stage ensures the creation of a complete reactor batch without material losses. According to the chosen configuration of reactors, one reactor batch can be processed daily.

However, at the same time, the reactor stage demonstrates the highest rate of utilization among all the stages of the process and the most extended waiting times. Because of the long processing time of one reactor batch, a queue occurs in front of the reactor because of its limited capacity. Nevertheless, the occurrence of waiting times does not cause any disruptions in the operation of the system or its blocking.

As far as the behavior of queues is concerned, it is essential to mention that they occur in front of the reactor and batching stages. On the contrary, the other stages show balance, effective performance, and absence of material supply issues.

Therefore, based on the simulation results, it can be said that the waste tire pyrolysis system will be capable of efficient and steady operation. The system is technically possible, whereas the reactor may act as a limiting factor because of its high productivity.

Table 9: *summarizes the overall evaluation of the major system performance indicators.*

Performance Indicator	Evaluation Result
Throughput Stability	Stable
Preprocessing Performance	Efficient
Reactor Utilization	Very High
Queue Formation	Mainly before reactor
Waiting Times	Acceptable under batch operation
System Congestion	Limited
Overall Productivity	Satisfactory
Main Limitation	Reactor cycle time

6. Chapter 6: Economic Feasibility Analysis

6.1. Cost Structure: CAPEX and OPEX

This chapter evaluates the economic feasibility of establishing a small-scale waste tire pyrolysis plant. The proposed system processes waste vehicle tires and converts them into valuable products such as diesel-like pyrolysis oil, carbon black, light fuel fraction, steel wires, and reusable gas.

The economic assumptions are connected to the simulation model developed in the previous chapters, where the system is designed around an average tire weight of **10 kg**, a daily input of **10–15 tons**, and a pyrolysis reactor capacity of **10 tons per batch**

Market-Based CAPEX Estimation

Diverse prices have been considered based on online sources for the cost of pyrolysis system for tires. **Pyrolysis systems capable of processing at a rate of 10 TPD** are available from suppliers ranging from 45,000 USD to 57,000 USD. The above prices generally do not include installation, utilities, safety measures, etc. Marketplaces provide two prices for 10 TPD plants, being 45,000 USD and average cost estimated at about 57,420 USD. Additional supplier information indicates that batch pyrolysis plants capable of producing 6-15 ton/day cost between 50,000 USD and 500,000 USD while for higher capacities, tire recycling plants cost in the range of 300,000 USD to 1,000,000 USD.

It is worth mentioning that online supplier prices generally only account for the central pyrolysis reactor or primary unit alone, whereas the capital expenditure required for an industrial facility entails not only the above factors but also those associated with the process prior to pyrolysis, such as preparation equipment, installation, infrastructure, etc.

From online source for currency converter and exchange rate between U.S. Dollars (USD) to Jordanian Dinar (JOD), it seems the approximate exchange rate for U.S. Dollars to Jordanian Dinners is **approximately 1 USD = 0.709 JOD**.

The cost of 700,000 JD thus seems quite reasonable considering that the cost includes the following (reactor, shredder/debreading machine, installation costs, utilities, emission control, safety measures, etc.).

Table 10: Estimated CAPEX Breakdown.

Item	Estimated Cost (JD)
Pyrolysis reactor / oven system	250,000
Tire debanding, cutting, and shredding line	120,000
Condensation and oil recovery system	70,000
Carbon black and steel wire handling	40,000
Gas treatment and safety system	45,000
Electrical, piping, and utilities	60,000
Installation, commissioning, and transport	50,000
Civil works and storage area	40,000
Contingency and miscellaneous costs	25,000
Total Estimated CAPEX	700,000 JD

Therefore, the final CAPEX of a complete waste tire pyrolysis facility is significantly higher than the standalone reactor price reported by online marketplaces, since a fully integrated plant requires additional industrial infrastructure, auxiliary systems, installation, and operational support units.

6.2. Operating Cost Structure

The operating cost includes labor, utilities, maintenance, daily machine operation, handling, and administrative costs. The project data assumes a monthly operating cost of **7,000–10,000 JD**, with 10–15 workers earning around **500 JD/month per worker**.

Table 11: *Estimated Monthly Operating Cost (OPEX) for the Waste Tire Pyrolysis Plant.*

Cost Item	Estimated Value
Workers	10–15 workers
Monthly wage per worker	500 JD
Labor cost	5,000–7,500 JD/month
Machine daily operating cost	150 JD/day
Maintenance, handling, and utilities	Included in monthly OPEX
Total monthly operating cost	7,000–10,000 JD/month

The daily equivalent operating cost is:

Table 12: *The daily equivalent operating cost.*

Scenario	Monthly OPEX	Daily OPEX
Low OPEX	7,000 JD/month	233 JD/day
High OPEX	10,000 JD/month	333 JD/day

6.3. Revenue Streams Analysis

The pyrolysis process produces multiple revenue streams. Literature and supplier sources commonly report that waste tire pyrolysis produces pyrolysis oil, recovered carbon black, steel wire, and combustible gas. Typical product yields are often around **40–45% oil**, **30–35% carbon black**, **10–15% steel wire**, and **5–10% syngas**, although the exact yield depends on tire type, reactor design, temperature, and operating conditions.

For this project, the assumed output distribution is:

Table 13: *The assumed output distribution.*

Product	Approximate Yield
Diesel / pyrolysis oil	40%
Carbon black	30–40%
Gas	10%
Light fuel fraction	10%
Steel wires	10%

The daily profit values provided for the project are:

Table 14: *The daily profit values.*

Product	Daily Profit Range (JD/day)
Diesel / pyrolysis oil	1,400–2,000
Carbon black	500
Light fuel fraction	100–150
Steel wires	150–200
Total Daily Product Profit	2,150–2,850

Gas is not treated as direct sales revenue in this model because it is usually reused as a heat source for the reactor. This reduces energy consumption and improves operating efficiency rather than generating direct cash income. Supplier sources also state that syngas can be reused as a heat source in the pyrolysis system.

6.4. Financial Modeling

The financial model is based on three operating scenarios:

Table 15: *The financial model is based on three operating scenarios.*

Scenario	Daily Product Profit	Daily OPEX	Net Daily Profit
Conservative	2,150 JD	333 JD	1,817 JD
Expected	2,500 JD	283 JD	2,217 JD
Optimistic	2,850 JD	233 JD	2,617 JD

The expected case is calculated using the midpoint of the daily profit range and the midpoint of the monthly operating cost.

6.5. Profitability Analysis

The expected monthly and annual net profits are:

Table 16: *The expected monthly and annual net profits.*

Scenario	Net Daily Profit	Monthly Net Profit	Annual Net Profit
Conservative	1,817 JD	54,510 JD	654,120 JD
Expected	2,217 JD	66,510 JD	798,120 JD
Optimistic	2,617 JD	78,510 JD	942,120 JD

These values indicate that the project can be financially attractive if daily production is stable and if product markets are available for pyrolysis oil, carbon black, and steel wires.

6.6. Break-even and Payback Period

The payback period is calculated as:

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Net Profit}}$$

Table 17: The payback period.

Scenario	Initial Investment	Annual Net Profit	Payback Period
Conservative	700,000 JD	654,120 JD	1.07 years
Expected	700,000 JD	798,120 JD	0.88 years
Optimistic	700,000 JD	942,120 JD	0.74 years

This indicates that the payback period for the project will be between **9 to 13 months** on the assumptions made. But it should be noted that this is an estimated financial calculation. The actual payback period can be extended because of factors such as licenses, maintenance, downtime, quality problems, and sales.

Previously, an economic analysis of a much smaller plant with the capacity to **process 3.5 tons** per day using the batch process indicated feasibility but had a higher payback period of **5 years**.

6.7. Return on Investment

ROI is calculated as:

$$ROI = \frac{\text{Annual Net Profit}}{\text{Initial Investment}} \times 100$$

Table 18: The expected ROI.

Scenario	ROI
Conservative	93.45%
Expected	114.02%
Optimistic	134.59%

The expected ROI is high because the daily profit assumptions are strong compared with the assumed CAPEX. This supports the feasibility of the project, but the model should still be treated as a preliminary feasibility study, not a final investment decision.

6.8. Sensitivity Analysis

The most important variables affecting feasibility are:

Table 19: *The most important variables affecting feasibility .*

Variable	Effect on Feasibility
Diesel / pyrolysis oil price	Very high impact
Daily production rate	Very high impact
Reactor downtime	High impact
Carbon black selling price	Medium to high impact
Labor and energy cost	Medium impact
CAPEX increase	Medium impact
Product quality	High impact

The reactor remains the most critical unit because it controls the total throughput of the plant. Since the reactor processes **10 tons per batch** and requires approximately **18–23 hours**, the project is highly sensitive to reactor availability and downtime.

6.9. Financial Sensitivity and Risk Analysis Using Python

However, it is important to develop a model to identify whether the waste tire pyrolysis plant project should continue or not. The model will perform a comprehensive sensitivity and risk analysis by evaluating the impact of increasing costs or decreasing product profit rates. Therefore, it will be much easier to make an informed decision since the model will analyze numerous possible outcomes of implementing this venture.

The following assumptions were made concerning investment and operation: the initial investment was set at 700,000 JD and the number of operational days per month was 30. The model considers three scenarios: conservative, expected, and optimistic. These scenarios involve changes in daily product profit rate and monthly cost rates under diverse economic conditions. In other words, this is a comprehensive approach allowing identifying possible consequences of increasing or reducing certain parameters in the economic market.

In the sensitivity section, there are several tests that determine the effect of raising or lowering such factors as operating cost rates and daily profit rates on the financial sustainability of the plant. For instance, it evaluates the potential outcome of a rise in operating costs by 10%, 20%, or even 50%. At the same time, the model performs the opposite analysis by estimating the effect of reducing daily profit rates by 10%, 20%, and so forth. All aspects of financial sustainability are considered, including higher labor rates, energy consumption, reduced prices, decreased demand, and so forth. It is evident that such a model helps identify the weak points of the project.

The results of calculations are provided for all possible combinations, and the following parameters are calculated: the new daily profit rates after changes, operating costs, net profits on a daily, monthly, and annual basis, payback period, and return on investment. Based on these figures, the project is categorized into four categories: Highly Feasible, Feasible, Risky, and Not Feasible. Such a classification simplifies decision-making since investors do not need to calculate everything manually.

The results are exported to a data table and stored in an Excel file for further consideration. In addition, the model generates a chart for the expected scenario. It depicts changes in monthly net profits in case of a decline in product profit rate and an increase in operating cost rates. Furthermore, the model defines the worst-case and best-case scenarios of the project. In summary, the implementation of this model facilitates evaluation.

The figures below show the Python code used to develop the financial analysis model, which was utilized to perform the sensitivity and risk analysis and to evaluate the economic feasibility of the proposed waste tire pyrolysis plant project as shown in figures below.

This code creates a financial feasibility model for a waste tire pyrolysis project. It defines the initial investment cost, operating days, and three business scenarios: conservative, expected, and optimistic. It also applies sensitivity analysis by testing different cost increases and profit decreases to see how these changes affect the project's profitability and risk.

```

import pandas as pd
import matplotlib.pyplot as plt

# -----
# Waste Tire Pyrolysis Feasibility Model
# -----

CAPEX = 700000 # Initial investment in JD
operating_days_per_month = 30

# Base scenarios
scenarios = {
    "Conservative": {"daily_product_profit": 2150, "monthly_opex": 10000},
    "Expected": {"daily_product_profit": 2500, "monthly_opex": 8500},
    "Optimistic": {"daily_product_profit": 2850, "monthly_opex": 7000}
}

# Sensitivity assumptions
cost_increases = [0, 0.10, 0.20, 0.30, 0.50]
profit_decreases = [0, 0.10, 0.20, 0.30, 0.40]

results = []

for scenario_name, data in scenarios.items():
    base_daily_profit = data["daily_product_profit"]
    base_monthly_opex = data["monthly_opex"]

    for cost_inc in cost_increases:
        for profit_dec in profit_decreases:

```

Figure 9: Python Code for Defining the Initial Assumptions and Financial Scenarios of the Waste Tire Pyrolysis Feasibility Model - 1.

```

        results.append({
            "Scenario": scenario_name,
            "Cost Increase": f"{int(cost_inc * 100)}%",
            "Profit Decrease": f"{int(profit_dec * 100)}%",
            "Net Daily Profit (JD)": round(net_daily_profit, 2),
            "Monthly Net Profit (JD)": round(monthly_net_profit, 2),
            "Annual Net Profit (JD)": round(annual_net_profit, 2),
            "Payback Period (Years)": round(payback_period, 2) if payback_period else "No Payback",
            "ROI (%)": round(roi, 2) if roi else "Negative",
            "Decision": decision
        })

df = pd.DataFrame(results)

# Display all results
print(df)

# Save results to Excel
df.to_excel("pyrolysis_feasibility_sensitivity_analysis.xlsx", index=False)

# -----
# Plot: Expected Scenario Sensitivity
# -----

expected_df = df[df["Scenario"] == "Expected"]

pivot_table = expected_df.pivot(
    index="Profit Decrease",
    columns="Cost Increase",

```

This part of the code stores the calculated financial results for each scenario in a results table. It includes the cost increase, profit decrease, net daily profit, monthly and annual net profit, payback period, ROI, and final decision. Then, the results are converted into a DataFrame, displayed, and saved as an Excel file. After that, the code filters the Expected scenario and prepares a pivot table to visualize the sensitivity analysis in a chart.

Figure 10: Python Code for Defining the Initial Assumptions and Financial Scenarios of the Waste Tire Pyrolysis Feasibility Model - 2.

```
adjusted_daily_profit = base_daily_profit * (1 - profit_dec)
adjusted_monthly_opex = base_monthly_opex * (1 + cost_inc)

daily_opex = adjusted_monthly_opex / operating_days_per_month
net_daily_profit = adjusted_daily_profit - daily_opex
monthly_net_profit = net_daily_profit * operating_days_per_month
annual_net_profit = monthly_net_profit * 12

if annual_net_profit > 0:
    payback_period = CAPEX / annual_net_profit
    roi = (annual_net_profit / CAPEX) * 100
else:
    payback_period = None
    roi = None

if annual_net_profit <= 0:
    decision = "Not Feasible"
elif payback_period <= 1.5:
    decision = "Highly Feasible"
elif payback_period <= 3:
    decision = "Feasible"
else:
    decision = "Risky"
```

This part of the code calculates the adjusted profit and operating costs after applying the sensitivity changes. It then finds the net daily, monthly, and annual profit. Based on the annual profit, the code calculates the payback period and ROI. Finally, it gives a decision for each case: Highly Feasible, Feasible, Risky, or Not Feasible depending on the project's profitability and payback period.

Figure 11: Python Code for Defining the Initial Assumptions and Financial Scenarios of the Waste Tire Pyrolysis Feasibility Model - 3.

```
results.append({
    "Scenario": scenario_name,
    "Cost Increase": f"{int(cost_inc * 100)}%",
    "Profit Decrease": f"{int(profit_dec * 100)}%",
    "Net Daily Profit (JD)": round(net_daily_profit, 2),
    "Monthly Net Profit (JD)": round(monthly_net_profit, 2),
    "Annual Net Profit (JD)": round(annual_net_profit, 2),
    "Payback Period (Years)": round(payback_period, 2) if payback_period else "No Payback",
    "ROI (%)": round(roi, 2) if roi else "Negative",
    "Decision": decision
})

df = pd.DataFrame(results)

# Display all results
print(df)

# Save results to Excel
df.to_excel("pyrolysis_feasibility_sensitivity_analysis.xlsx", index=False)

# -----
# Plot: Expected Scenario Sensitivity
# -----

expected_df = df[df["Scenario"] == "Expected"]

pivot_table = expected_df.pivot(
    index="Profit Decrease",
    columns="Cost Increase",
```

This part of the code saves all the financial analysis results into a table. It records the scenario name, cost increase, profit decrease, net profit, payback period, ROI, and final decision. Then, it converts the results into a DataFrame, displays them, and exports them to an Excel file. Finally, it filters the **Expected** scenario and prepares a pivot table to create a sensitivity analysis chart.

Figure 12: Python Code for Defining the Initial Assumptions and Financial Scenarios of the Waste Tire Pyrolysis Feasibility Model - 4.

```
plt.figure(figsize=(10, 6))

for column in pivot_table.columns:
    plt.plot(
        pivot_table.index,
        pivot_table[column],
        marker="o",
        linewidth=2,
        label=f"Cost Increase {column}"
    )

plt.title("Sensitivity Analysis: Monthly Profit under Cost Increase and Profit Decrease")
plt.xlabel("Profit Decrease")
plt.ylabel("Monthly Net Profit (JD)")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()

# -----
# Worst Case Summary
# -----

worst_case = df.sort_values("Annual Net Profit (JD)").head(1)
best_case = df.sort_values("Annual Net Profit (JD)").tail(1)

print("\nWorst Case:")
print(worst_case)

print("\nBest Case:")
print(best_case)
```

This part of the code creates a sensitivity analysis chart for the Expected scenario. It plots how the monthly net profit changes when profit decreases and costs increase. The chart helps show the financial risk visually. After that, the code identifies the worst case and best case results based on annual net profit, then prints both cases for comparison.

Figure 13: Python Code for Defining the Initial Assumptions and Financial Scenarios of the Waste Tire Pyrolysis Feasibility Model - 5.

The findings of the financial sensitivity analysis for the waste tire pyrolysis project **are shown in Table below**. It reflects the changes of project performance when costs of operation increase and product profit rate decreases in conservative, expected, and optimistic cases. **In all the scenarios, the net profit per day, monthly and yearly net profits, payback period, and ROI** are calculated to classify the project as highly

feasible, feasible, risky, and not feasible respectively. It is important to note that the project shows its economic viability under various conditions tested by the financial sensitivity analysis.

Table 20: Financial Sensitivity Analysis Results for the Waste Tire Pyrolysis Plant under Different Cost Increase and Profit Decrease Scenarios.

Scenario	Cost Increase	Profit Decrease	Net Daily Profit (JD)	Monthly Net Profit (JD)	Annual Net Profit (JD)	Payback Period (Years)	ROI (%)	Decision
Conservative	0%	0%	1816.67	54500	654000	1.07	93.43	Highly Feasible
Conservative	0%	10%	1601.67	48050	576600	1.21	82.37	Highly Feasible
Conservative	0%	20%	1386.67	41600	499200	1.4	71.31	Highly Feasible
Conservative	0%	30%	1171.67	35150	421800	1.66	60.26	Feasible
Conservative	0%	40%	956.67	28700	344400	2.03	49.2	Feasible
Conservative	10%	0%	1783.33	53500	642000	1.09	91.71	Highly Feasible
Conservative	10%	10%	1568.33	47050	564600	1.24	80.66	Highly Feasible
Conservative	10%	20%	1353.33	40600	487200	1.44	69.6	Highly Feasible
Conservative	10%	30%	1138.33	34150	409800	1.71	58.54	Feasible
Conservative	10%	40%	923.33	27700	332400	2.11	47.49	Feasible
Conservative	20%	0%	1750	52500	630000	1.11	90	Highly Feasible
Conservative	20%	10%	1535	46050	552600	1.27	78.94	Highly Feasible
Conservative	20%	20%	1320	39600	475200	1.47	67.89	Highly Feasible
Conservative	20%	30%	1105	33150	397800	1.76	56.83	Feasible
Conservative	20%	40%	890	26700	320400	2.18	45.77	Feasible
Conservative	30%	0%	1716.67	51500	618000	1.13	88.29	Highly Feasible
Conservative	30%	10%	1501.67	45050	540600	1.29	77.23	Highly Feasible
Conservative	30%	20%	1286.67	38600	463200	1.51	66.17	Feasible
Conservative	30%	30%	1071.67	32150	385800	1.81	55.11	Feasible
Conservative	30%	40%	856.67	25700	308400	2.27	44.06	Feasible
Conservative	50%	0%	1650	49500	594000	1.18	84.86	Highly Feasible

Conservative	50%	10%	1435	43050	516600	1.36	73.8	Highly Feasible
Conservative	50%	20%	1220	36600	439200	1.59	62.74	Feasible
Conservative	50%	30%	1005	30150	361800	1.93	51.69	Feasible
Conservative	50%	40%	790	23700	284400	2.46	40.63	Feasible
Expected	0%	0%	2216.67	66500	798000	0.88	114	Highly Feasible
Expected	0%	10%	1966.67	59000	708000	0.99	101.14	Highly Feasible
Expected	0%	20%	1716.67	51500	618000	1.13	88.29	Highly Feasible
Expected	0%	30%	1466.67	44000	528000	1.33	75.43	Highly Feasible
Expected	0%	40%	1216.67	36500	438000	1.6	62.57	Feasible
Expected	10%	0%	2188.33	65650	787800	0.89	112.54	Highly Feasible
Expected	10%	10%	1938.33	58150	697800	1	99.69	Highly Feasible
Expected	10%	20%	1688.33	50650	607800	1.15	86.83	Highly Feasible
Expected	10%	30%	1438.33	43150	517800	1.35	73.97	Highly Feasible
Expected	10%	40%	1188.33	35650	427800	1.64	61.11	Feasible
Expected	20%	0%	2160	64800	777600	0.9	111.09	Highly Feasible
Expected	20%	10%	1910	57300	687600	1.02	98.23	Highly Feasible
Expected	20%	20%	1660	49800	597600	1.17	85.37	Highly Feasible
Expected	20%	30%	1410	42300	507600	1.38	72.51	Highly Feasible
Expected	20%	40%	1160	34800	417600	1.68	59.66	Feasible
Expected	30%	0%	2131.67	63950	767400	0.91	109.63	Highly Feasible
Expected	30%	10%	1881.67	56450	677400	1.03	96.77	Highly Feasible
Expected	30%	20%	1631.67	48950	587400	1.19	83.91	Highly Feasible
Expected	30%	30%	1381.67	41450	497400	1.41	71.06	Highly Feasible
Expected	30%	40%	1131.67	33950	407400	1.72	58.2	Feasible
Expected	50%	0%	2075	62250	747000	0.94	106.71	Highly Feasible
Expected	50%	10%	1825	54750	657000	1.07	93.86	Highly Feasible

Expected	50%	20%	1575	47250	567000	1.23	81	Highly Feasible
Expected	50%	30%	1325	39750	477000	1.47	68.14	Highly Feasible
Expected	50%	40%	1075	32250	387000	1.81	55.29	Feasible
Optimistic	0%	0%	2616.67	78500	942000	0.74	134.57	Highly Feasible
Optimistic	0%	10%	2331.67	69950	839400	0.83	119.91	Highly Feasible
Optimistic	0%	20%	2046.67	61400	736800	0.95	105.26	Highly Feasible
Optimistic	0%	30%	1761.67	52850	634200	1.1	90.6	Highly Feasible
Optimistic	0%	40%	1476.67	44300	531600	1.32	75.94	Highly Feasible
Optimistic	10%	0%	2593.33	77800	933600	0.75	133.37	Highly Feasible
Optimistic	10%	10%	2308.33	69250	831000	0.84	118.71	Highly Feasible
Optimistic	10%	20%	2023.33	60700	728400	0.96	104.06	Highly Feasible
Optimistic	10%	30%	1738.33	52150	625800	1.12	89.4	Highly Feasible
Optimistic	10%	40%	1453.33	43600	523200	1.34	74.74	Highly Feasible
Optimistic	20%	0%	2570	77100	925200	0.76	132.17	Highly Feasible
Optimistic	20%	10%	2285	68550	822600	0.85	117.51	Highly Feasible
Optimistic	20%	20%	2000	60000	720000	0.97	102.86	Highly Feasible
Optimistic	20%	30%	1715	51450	617400	1.13	88.2	Highly Feasible
Optimistic	20%	40%	1430	42900	514800	1.36	73.54	Highly Feasible
Optimistic	30%	0%	2546.67	76400	916800	0.76	130.97	Highly Feasible
Optimistic	30%	10%	2261.67	67850	814200	0.86	116.31	Highly Feasible
Optimistic	30%	20%	1976.67	59300	711600	0.98	101.66	Highly Feasible
Optimistic	30%	30%	1691.67	50750	609000	1.15	87	Highly Feasible
Optimistic	30%	40%	1406.67	42200	506400	1.38	72.34	Highly Feasible

Optimistic	50%	0%	2500	75000	900000	0.78	128.57	Highly Feasible
Optimistic	50%	10%	2215	66450	797400	0.88	113.91	Highly Feasible
Optimistic	50%	20%	1930	57900	694800	1.01	99.26	Highly Feasible
Optimistic	50%	30%	1645	49350	592200	1.18	84.6	Highly Feasible
Optimistic	50%	40%	1360	40800	489600	1.43	69.94	Highly Feasible

The Figure below is a representation of the variation in monthly net profit of the plant in response to the expected scenario of decreased product profit and increased operating cost. In the diagram, the X-axis is denoted as a percentage decrease in product profit, while the Y-axis denotes the monthly net profit in Jordanian Dinars (JD). The curves represent the level of increase in operating costs, which varies between 0% and 50%. As can be observed from the figure below, there is an obvious trend in which monthly net profit decreases with decreasing product profit and increasing operating costs. Despite these scenarios, the project still yields positive monthly profits at all levels.

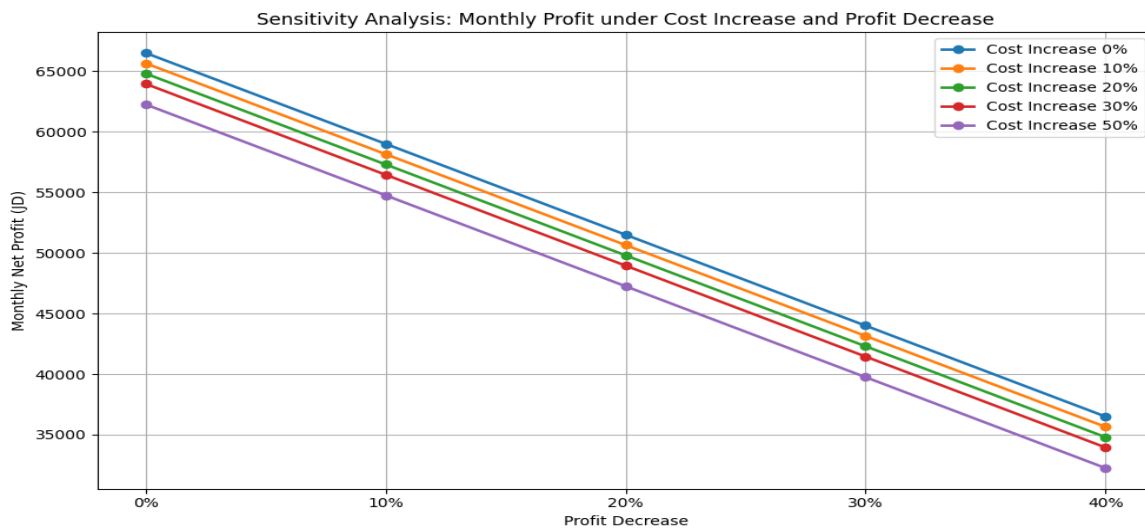


Figure 14: Sensitivity Analysis of Monthly Net Profit under Different Cost Increase and Profit Decrease Scenarios.

6.10. Final Economic Feasibility Decision

This Python code is simple, short, and useful. It calculates daily profit, monthly profit, annual profit, payback period, and ROI for the three scenarios.

```

# Economic Feasibility Model for Waste Tire Pyrolysis Plant

capex = 700000 # JD

scenarios = {
    "Conservative": {"daily_profit": 2150, "monthly_opex": 10000},
    "Expected":      {"daily_profit": 2500, "monthly_opex": 8500},
    "Optimistic":    {"daily_profit": 2850, "monthly_opex": 7000}
}

for name, data in scenarios.items():
    daily_opex = data["monthly_opex"] / 30
    net_daily_profit = data["daily_profit"] - daily_opex
    monthly_net_profit = net_daily_profit * 30
    annual_net_profit = monthly_net_profit * 12
    payback_period = capex / annual_net_profit
    roi = (annual_net_profit / capex) * 100

    print(f"\n{name} Scenario")
    print("-" * 30)
    print(f"Daily Product Profit: {data['daily_profit']:.2f} JD/day")
    print(f"Daily OPEX: {daily_opex:.2f} JD/day")
    print(f"Net Daily Profit: {net_daily_profit:.2f} JD/day")
    print(f"Monthly Net Profit: {monthly_net_profit:.2f} JD/month")
    print(f"Annual Net Profit: {annual_net_profit:.2f} JD/year")
    print(f"Payback Period: {payback_period:.2f} years")
    print(f"ROI: {roi:.2f}%")

```

This code calculates the economic feasibility of the waste tire pyrolysis plant using three scenarios: conservative, expected, and optimistic. It uses the initial investment cost, daily profit, and monthly operating expenses to calculate the net daily profit, monthly profit, annual profit, payback period, and ROI. The results help compare the project's financial performance under different conditions and show whether the investment is profitable.

Figure 15: Economic Feasibility Model for Waste Tire Pyrolysis Plant.

The program calculates three possibilities in terms of finance. The conservative assumption has lower daily profit and higher operating costs. The expected is the most realistic possibility in the middle. The optimistic case has higher profits in sales but lower operating costs.

This makes the program helpful because it does not give a single definitive answer. It demonstrates how the feasibility changes based on operating conditions.

Based on the market price searches and project assumptions, the proposed waste tire pyrolysis plant is economically feasible. The assumed total cost of the project at 700,000 JD is reasonable since it represents a complete installed line, albeit higher than the basic online machine price since it does not include installation costs, pre-processing of materials, safety devices, civil works, and installation costs.

The project will be profitable due to multiple revenue streams from the output, such as diesel-like pyrolysis oil, carbon black, light fuel, and steel wires. The assumed net income per day is around 2,217 JD/day, translating into a projected net income of 798,120 JD/year.

Based on the project assumptions, the expected payback period would be around 0.88 years, while the expected return on investment (ROI) is approximately 114%.

Nevertheless, the final investment decision shall depend on other factors like the quality of the output products, environmental clearance, emission control, marketing arrangements for the carbon black and pyrolysis oil, the frequency of reactor downtime, and maintenance costs.

7

Chapter 7: Environmental Impact Analysis

7.1. Environmental Assessment Framework

In this chapter, the potential environmental impact of the proposed small-scale waste tire pyrolysis plant will be analyzed. In terms of environmental assessment, the major effects to be considered are those related to processing waste tires to generate valuable products, including pyrolysis oil, carbon black, steel wires, light fuel fraction, and gas. This is why the waste tire conversion system can be evaluated as waste-to-resource conversion, where the accumulated waste is reduced and secondary materials and energy products are generated (Han et al., 2023; Wu et al., 2025).

The framework for environmental assessment consists in comparing the proposed pyrolysis technology with the traditional approaches to waste tire treatment, such as landfilling, open burning, uncontrolled burning, and tire combustion. The latter two options have more negative consequences than waste tires recycling or recovery because they involve higher consumption of new materials, and thus, generate an increased environmental footprint compared to recycling and recovery (U.S. Environmental Protection Agency [EPA], 2019). It is indicated that, according to the US EPA Waste Reduction Model, tire recycling contributes to greater GHG emissions reduction than combustion and landfilling due to saving energy for virgin secondary material production.

The proposed approach is to be estimated in terms of environmental impact related to waste reduction, carbon emissions, avoided emissions, energy recovery, material recovery, etc. In this case, the processing capacity of the system amounts to about 10-15 tons per day with a tire weight equal to 10 kg and 10 tons per batch capacity of the pyrolysis reactor.

7.2. Carbon Emissions Estimation

The CO₂ emissions from the plant depend on reactor heating, electrical energy, material transport, and machinery operations. The environmental accounting system often calculates emissions by using a simple equation where emissions are equal to activity levels multiplied by emission factors. Thus, the formula for estimating carbon emissions is:

$$CO_2 \text{ Emissions} = \text{Activity Data} \times \text{Emission Factor}$$

For instance, if the fuel used as the external source is diesel, then the daily carbon emissions can be calculated by multiplying the amount of diesel fuel with the factor for emissions of diesel. Nevertheless, waste tire pyrolysis technology has the environmental benefit since the gas emitted at this stage can be reused as a source of heat energy for the reactor. Thus, the energy efficiency of this process is increased significantly (Han et al., 2023; Wu et al., 2025).

In the proposed system, there are several sources of emissions that should be analyzed to calculate the environmental benefits. These include reactor heating, electricity use, pre-processing equipment, transportation of materials and products. The reactor is considered the most significant in terms of emissions since it works continuously and consumes high amounts of energy.

7.3. Carbon Savings and Avoided Emissions

Carbon emissions reduction takes place through the utilization of pyrolysis outputs as a substitute for traditional materials or through the redirection of waste tires from environmentally unfriendly disposal channels. For this project, the key avoided emission opportunities include avoidance of waste tires landfills, substitution of a fraction of fossil-derived fuels by pyrolysis liquids, substitution of virgin carbon black by recycled carbon black, and recovery of steel wires rather than just using new steel extraction.

According to the Environmental Protection Agency (EPA), tire recycling yields higher carbon emissions mitigation benefits than combustion or disposal in landfills due to lower energy use for the manufacture of secondary products (EPA, 2019). This strengthens the environmental rationale of the suggested project since waste tires are used not as waste, but as raw materials for producing recyclable materials.

Table 21: *The proposed distribution of the system output.*

Product	Approximate Yield
Diesel / pyrolysis oil	40%
Carbon black	30–40%
Gas	10%
Light fuel fraction	10%
Steel wires	10%

For a daily processing capacity of 10 tons, the approximate recovered quantities are:

Table 22: *Daily processing capacity of 10 tons.*

Product	Estimated Quantity per 10 tons/day
Pyrolysis oil / diesel-like product	4.0 tons/day
Carbon black	3.0–4.0 tons/day
Gas	1.0 ton/day equivalent
Light fuel fraction	1.0 ton/day
Steel wires	1.0 ton/day

These recovered products reduce the environmental burden of waste tires and support circular economy principles. Several reviews confirm that waste tire pyrolysis can recover oil, gas, carbon black, and steel wire, but the environmental benefit depends on product quality, upgrading, and final application (**Han et al., 2023; Wu et al., 2025**).

7.4. Energy Recovery Impact

Energy recovery is one of the most important environmental benefits of waste tire pyrolysis. The process produces pyrolysis oil and non-condensable gas, both of which have energy value. The gas fraction can be reused internally to support reactor heating, while pyrolysis oil can be used as an industrial fuel or upgraded into a diesel-like product.

According to **Han et al. (2023)**, waste tire pyrolysis produces several useful products, including oil, gas, black carbon, and steel wire. From an environmental perspective, the reuse of gas is particularly important because it reduces external fuel demand and improves thermal efficiency. **Wu et al. (2025)** also discusses the carbon mitigation potential of waste tire pyrolysis from a life cycle perspective, especially when recovered products substitute conventional fuels and materials.

In this project, energy recovery contributes to environmental performance through three pathways:

Table 23: *Energy recovery contributes to environmental performance through three pathways.*

Energy Recovery Pathway	Environmental Contribution
Reuse of pyrolysis gas	Reduces external fuel consumption
Use of pyrolysis oil	Provides alternative industrial fuel
Heat integration	Improves process efficiency and reduces emissions

Therefore, energy recovery improves both the environmental and economic performance of the system.

7.5. Sustainability Assessment

Sustainability in the suggested pyrolysis plant can be examined through its environmental, economic, and operational aspects. Environmentally, it will address the issue of accumulating waste tires and transform them into valuable products. Economically, the plant generates revenue from several sources, including pyrolysis oil, carbon black, steel wires, and fuel fraction. From an operational standpoint, the system requires dependable reactor operation and appropriate management of preprocessing, batching, and product separation.

Tires are regarded as one of the significant environmental issues, owing to their non-degradable nature and the potential dangers associated with their disposal, such as fire hazards, storage risks, and pollution. Pyrolysis is extensively covered in scholarly articles as a potentially viable thermally driven method since it transforms waste tires into energy and material products (Han et al., 2023; Nkosi et al., 2020).

The sustainability assessment is summarized below:

Table 24: *The sustainability assessment.*

Sustainability Dimension	Positive Contribution
Waste management	Reduces tire accumulation
Resource recovery	Produces oil, carbon black, steel, and gas
Energy efficiency	Reuses gas for heating

Circular economy	Converting waste into marketable products
Economic feasibility	Generates multiple revenue streams

Overall, the proposed system supports sustainability by combining waste reduction, energy recovery, and material recovery.

7.6. Environmental Trade-offs

Although waste tire pyrolysis provides several environmental benefits, it also has environmental trade-offs. First, the process requires high temperature, which means that energy consumption can be significant if the plant depends heavily on external fossil fuel. Second, pyrolysis oil may contain sulfur and other compounds that require upgrading before use in sensitive applications. Third, recovered carbon black may need further treatment before replacing virgin carbon black in high-value applications. Therefore, product quality and emission control are essential for achieving strong environmental performance (Han et al., 2023; Wu et al., 2025).

Another important trade-off is that the pyrolysis plant must be operated under controlled industrial conditions. Without proper gas cleaning, condensation, odor control, and safety systems, the environmental benefit may decrease. Therefore, the proposed project should include emission treatment, environmental permits, product testing, and continuous monitoring.

Table 25: *Environmental Trade-offs.*

Environmental Benefit	Possible Trade-off
Reduces waste tire disposal	Requires controlled industrial operation
Recovers fuel and materials	Product quality may require upgrading
Reuses pyrolysis gas	External energy may still be needed
Products recovered carbon black	May require post-treatment
Avoids landfill accumulation	Requires emission control and permits

7.7. Discussion of Environmental Results

The environmental analysis indicates that the proposed waste tire pyrolysis plant has strong environmental potential when properly designed and controlled. The main benefit is the conversion of waste tires into useful products instead of sending them to landfills, open dumping areas, or uncontrolled burning. This supports circular economy principles and reduces the environmental burden associated with end-of-life tires.

The system also provides material and energy recovery benefits. Pyrolysis oil and gas can contribute to energy recovery, while recovered carbon black and steel wires can reduce the need for virgin raw materials. These benefits are consistent with previous studies showing that waste tire pyrolysis can be a feasible waste-to-resource pathway when environmental control, product upgrading, and stable markets are considered (Han et al., 2023; Nkosi et al., 2020; Wu et al., 2025).

However, the final environmental performance depends on practical implementation. The plant must include emission control systems, safe operation procedures, product quality testing, and proper handling of sulfur-containing compounds. Therefore, the project can be considered environmentally promising, but only if it is implemented with adequate environmental safeguards and regulatory compliance.

8. Chapter 8: Proposed BioFuel Mobile Application Interface

8.1. Concept of the Application

As an additional practical contribution to the project, a simple mobile application interface was proposed under the name BioFuel. The application is designed as a front-end prototype that supports the monitoring of the waste tyre pyrolysis system. It does not represent a fully connected industrial control system at this stage; rather, it demonstrates how the operational data of the plant can be presented in a clear, visual, and decision-supporting format.

The proposed interface is linked to the main logic of the simulation model, where the plant consists of several connected operations such as debinding, cutting, shredding, buffering, batching, pyrolysis, and product separation. Since the reactor is the most critical unit in the system and represents the main bottleneck, the application gives special attention to reactor status, material level, temperature, humidity, batch readiness, and maintenance needs.

8.2. Application Objectives

The main objective of the BioFuel mobile interface is to improve operational visibility by allowing the user to monitor the plant condition through a simple dashboard. The application also aims to support faster decision-making by showing machine status, material availability, reactor loading percentage, alerts, and basic production reports.

The application concept supports the project by transforming simulation and feasibility results into a practical monitoring idea. Instead of presenting the plant only as a simulation model, the mobile interface shows how the same operational logic can be used in a real industrial environment for tracking, control support, and preventive maintenance planning.

8.3. General Dashboard Interface

The first screen of the application is a general dashboard that summarizes the overall status of the BioFuel plant. It provides a quick view of daily capacity, system status, available material, active alerts, and the condition of each major machine. The dashboard includes the debinding machine, cutting machine, shredding machine, and pyrolysis reactor.

For each machine, the interface displays the current operating condition, such as Good, Bad, High Load, or Needs Maintenance. It also presents the amount of material currently being handled, machine temperature,

humidity level, and operation load. This makes the dashboard useful for identifying abnormal behavior before it causes production delay.

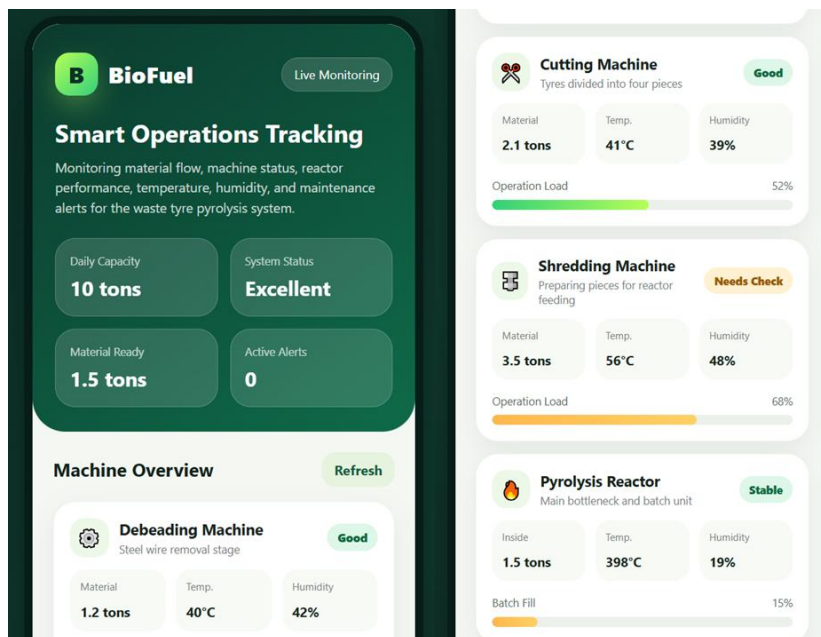


Figure 16: General Dashboard Interface.

8.4. Reactor-Specific Interface

The second major screen is dedicated to the pyrolysis reactor. This screen is important because the reactor controls the total throughput of the system. The reactor interface shows the current material inside the reactor, the required amount for a full batch, the remaining material needed, reactor temperature, humidity, estimated time left, and maintenance status.

The reactor screen also includes a circular batch-fill indicator. This indicator shows the percentage of reactor loading and helps the operator understand whether the reactor is still waiting for material, operating under high utilization, or ready to start a new batch. Since one reactor batch requires approximately 10 tons of material, the visual indicator makes the batch condition easier to interpret.

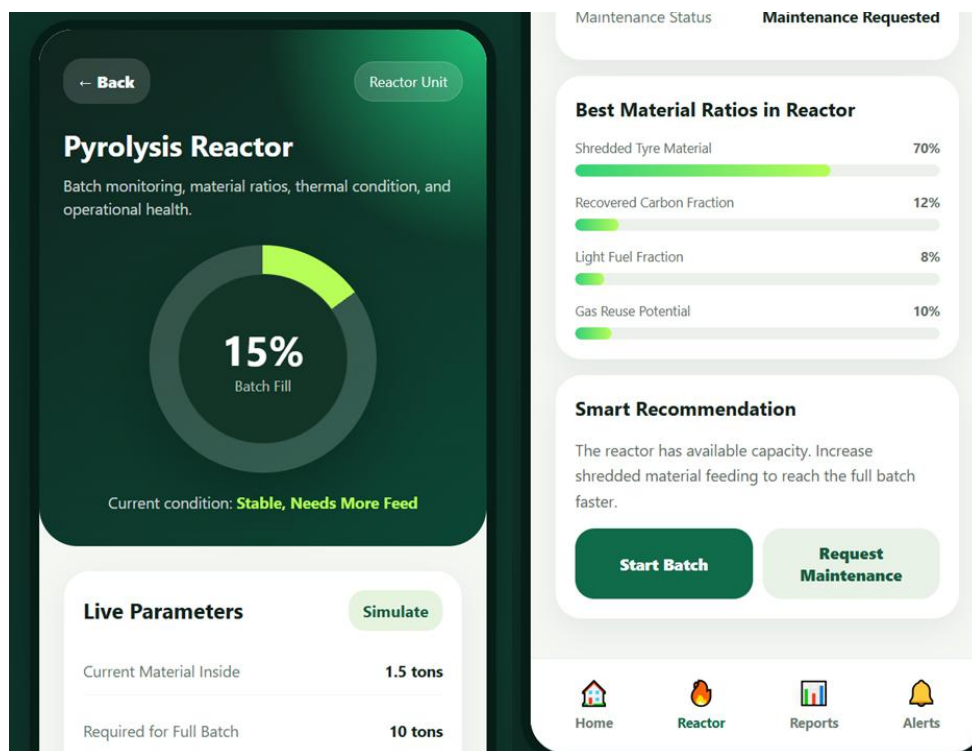


Figure 17: Reactor-Specific Interface.

8.5. Monitoring Parameters Included in the Application

Table 26: *Monitoring Parameters Included in the Application.*

Monitoring Element	Displayed Data	Purpose
Machine Status	Good, Bad, High Load, Needs Maintenance	To identify operational problems quickly.
Material Level	Current material quantity and required material	To track feeding and batching readiness.
Temperature	Machine and reactor temperature	To monitor thermal condition and safety.
Humidity	Humidity percentage around the process	To support environmental and operating condition tracking.
Reactor Fill Level	Percentage of batch completion	To show whether the reactor is ready for operation.
Alerts	Critical warnings and maintenance notifications	To support preventive action and reduce downtime.
Reports	Daily, weekly, and monthly production summaries	To evaluate throughput, utilization, and profitability.

8.6. Reports and Alerts Features

The application includes a reports screen that presents simplified operational results. The report can show daily, weekly, and monthly summaries, such as processed material, completed batches, average reactor utilization, and estimated net profit. This feature links the operational side of the project with the economic feasibility analysis, because production stability and reactor utilization directly affect profitability.

The alerts screen provides warnings related to high reactor utilization, shredding machine temperature, buffer accumulation, and maintenance needs. Each alert can be marked as resolved, and the interface can display suggested actions such as viewing the reactor screen or generating a maintenance work order. This feature supports preventive maintenance and reduces the risk of unplanned downtime.

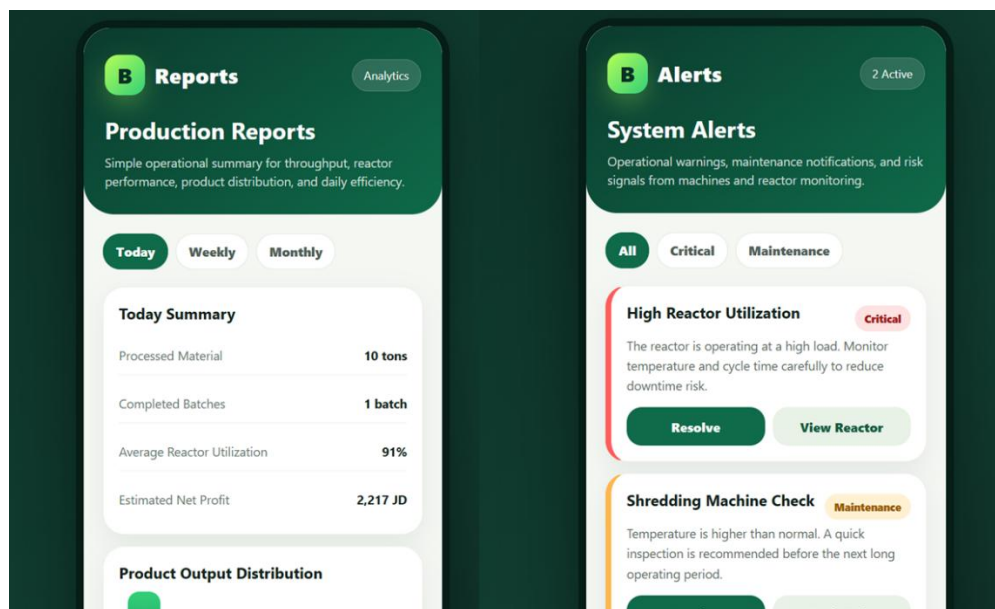


Figure 18: Reports and Alerts Features.

8.7. Interactive Features of the Prototype

The prototype is designed to be more than a static screen. It includes simple interactive functions such as refreshing live data, simulating reactor filling, requesting maintenance, starting a batch when the reactor is ready, filtering alerts, and switching between report periods. These features demonstrate how a future full application could help operators interact with plant data in real time.

8.8. Added Value to the Project

Adding the BioFuel mobile application concept improves the practical value of the project. The original study evaluates the plant technically, economically, and environmentally, while the application adds an

operational management layer. This means that the proposed project is not only a feasible pyrolysis system, but also a system that can be monitored and managed through a smart digital interface.

The application also strengthens the connection between simulation and industrial implementation. The simulation identifies the reactor as the main bottleneck, and the mobile interface translates this finding into a practical monitoring feature by focusing on reactor load, temperature, batch readiness, and maintenance condition. Therefore, the application supports better process control, faster response to problems, and improved operational reliability.

8.9.Future Development of the Application

In future work, the BioFuel interface can be developed from a front-end prototype into a complete monitoring system. This can be achieved by connecting the application to real sensors, machine controllers, databases, and cloud dashboards. Temperature sensors, humidity sensors, weight sensors, and machine status signals can provide live data to the application.

A future version may also include artificial intelligence features, such as predicting reactor downtime, recommending the best feeding strategy, estimating product output, and warning the operator before machine failure occurs. In this way, the application can become a smart decision-support tool for waste tyre pyrolysis plant management.

9. Chapter 9: Discussion

9.1. Integrated Analysis: Simulation, Economic, and Environmental

It is clear from the results of the current research that the proposed small-scale waste tyre pyrolysis system is better assessed considering all dimensions – simulation, economic, and environmental. Every aspect provides its unique point on the feasibility of the project.

In terms of simulation analysis, the proposed system is technically feasible for the processing of waste tyres. The system's flow includes the following elements: debanding, cutting, shredding, buffering, batching, pyrolysis, and product separation. Simulation results suggest that the system's preprocessing phases will provide materials for processing continuously. Nevertheless, reactor will remain the critical bottleneck since it will operate in batch mode with rather lengthy processing times (about 18-23 hours).

Speaking about economic feasibility, it is necessary to say that the project is expected to be profitable provided that the conditions assumed are correct. It is planned that the total amount of capital invested will make 700,000 JD. Daily net profit will be around 2,217 JD/day. Thus, the expected annual net profit is about 798,120 JD, which means that the project can pay back within 0.88 years. In other words, the project can be seen as a potentially viable enterprise.

Environmentally speaking, it is important to mention that the project can be environmentally efficient. In order not to burden landfills or burn waste tyres uncontrolledly, they are to be processed into various products: pyrolysis oil, carbon black, steel wires, gas, and light fuel fraction. From this point of view, the project follows the principle of the circular economy since waste tyres will become sources of valuable materials and energy.

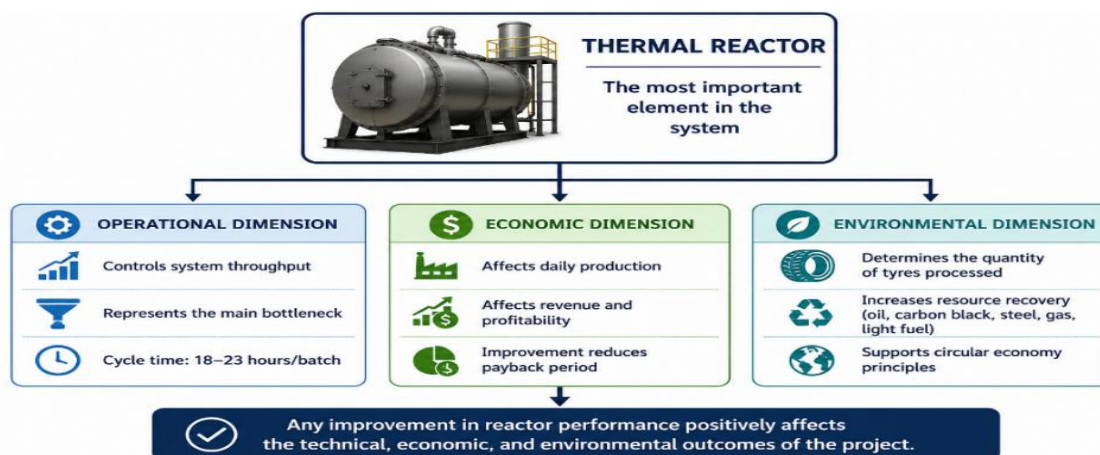


Figure 19: The Central Role of the Reactor in the Integrated Analysis of the Waste Tyre Pyrolysis System.

It is possible to state that reactor remains the most important component of the entire system under discussion. Indeed, it operates as a limiting factor in terms of operation, economic efficiency, and environmental effect since it will define both throughput, daily profits/losses, and the degree of waste processing.

9.2. Comparison with Literature

The results obtained from this study coincide with those in the previous literature, which recognize waste tyre pyrolysis as an economically beneficial approach to waste recycling. Several papers have indicated that tyre pyrolysis technology can transform end-of-life tires into various useful resources like oil, gas, carbon black, and steel wires. The results of this study provide evidence to support this finding since they demonstrate the capability of a small-scale plant to produce revenues in different areas.

Regarding the output assumptions, they comply with the yield distribution range described in many literature pieces on waste tyre pyrolysis. The considered yield distribution includes about 40% of pyrolysis oil, 30-40% carbon black, 10% gas, 10% light fuel fraction, and 10% steel wires. They correspond to the typical yield distribution range in a variety of studies, which implies that a small-scale plant can be productive and generate numerous resources.

As far as the simulation is concerned, many publications on tyre pyrolysis focus on thermodynamics, chemical reaction modeling, products' compositions, and process optimization. On the contrary, this paper provides information on the application of discrete-event simulation to investigate flow and batching issues, queues, utilization of a particular machine, bottlenecks, and other features of a small-scale plant operation. It is necessary due to the nature of the studied process, which implies both chemical process and production system.

When compared with previous economic analysis results, the calculated payback period appears rather short, even though this study uses some assumptions concerning product prices, running costs, equipment costs, and availability. For instance, previous literature indicated much longer payback periods at the same plant scale and capacity. Thus, it can be concluded that economic performance heavily depends on the specified parameters.

Therefore, the paper finds itself agreeing with literature on technical and environmental aspects of waste tyre pyrolysis, but not with previous economic analysis methods.

9.3. Practical Implications

The implications for practical application are evident from this analysis. It is clear that a mini waste tyre pyrolysis plant can serve as a promising investment option in case it is properly designed, operated, and possesses reliable market outlets for the produced goods.

To potential investors, the preliminary economic indicators reveal that this project can be profitable enough. The estimated payback period is lower than one year in the considered scenario. Nevertheless, it does not guarantee positive net present value due to product selling prices, reliability of output generation, maintenance costs, fuel expenses, product quality, and governmental regulations.

For plant designers, it can be stated that the reactor needs to be focused upon in its optimization process because it is recognized as the principal bottleneck. Consequently, the efforts can be concentrated on such factors as reactor capacity, cycle time, loading/unloading time, energy saving, and maintenance. The number of reactors can be increased, or the duration of cycle time can be shortened to boost the capacity.

From the point of view of waste management authorities, this project provides a solution for dealing with the environmental problem caused by waste tyres. This approach ensures sustainable waste management and implementation of the circular economy model.

Finally, to plant operators, the simulation results indicate the importance of buffer management and batching policy selection. As is known, the reactor operates only if there is a complete batch in it.

9.4. Limitations of the Study

However, there are several shortcomings that need to be mentioned despite the valuable information obtained.

In the first place, the simulation model relies on assumptions regarding its input parameters such as processing times, capacities, and other characteristics. In a real plant, all of them may be different due to the machinery used, tyre properties, personnel proficiency, and other factors.

The second limitation concerns the lack of precise modeling of the chemical reactions. The pyrolysis stage is defined as a batch process with its time distribution, however, the reactions themselves, changes in temperature, duration effect, catalysts' role, etc., are not taken into account.

As far as the economic analysis is concerned, it also requires some assumptions since the value of revenue and costs is unknown. Thus, product prices will vary depending on the current demand, their quality, acceptability, etc. The results of this research are only preliminary estimates.

The fifth limitation concerns the environmental aspect. Although environmental analysis is performed, it is conducted at the theoretical level, which means that no complete life cycle assessment has been done. Consequently, carbon emission reduction, energy balance, and others cannot be calculated accurately.

Finally, the model supposes a stable operation of the system; hence, it may be disrupted by downtime, maintenance, start-up issues, product quality, labor problems, and so on.

10.

Chapter 10: Conclusion and Recommendations

10.1. Summary of Findings

In this study, the technical and economic feasibility of the small-scale tyre pyrolysis process has been assessed using an integrative approach consisting of discrete-event simulation, feasibility analysis, and environmental assessment. The discrete event simulation model was built to simulate the different phases of the process, which include tyre receiving, deveaning, cutting, shredding, buffer storage, batching, pyrolysis, and product separation.

According to the simulation results, the pre-processing phases of the process will work effectively and supply materials continuously to the batching phase. In addition, it has been found out that the main bottleneck of the process is the pyrolysis phase since it is operated using batch method and requires a relatively long process time of about 18–23 h per batch. The batch size of the reactor equals 10 tons, which results in only one batch being processed for one day.

Based on the economic analysis carried out, it can be concluded that the studied project may have good financial feasibility since according to the calculations the initial capital investment will equal 700,000 JD, the daily net income will amount to about 2,217 JD/day, while the annual net income will be approximately 798,120 JD. Furthermore, the expected payback period will be 0.88 years, and the ROI will be about 114%.

It was shown by environmental analysis that the process under investigation can contribute to sustainable waste management due to transformation of waste tyres into useful products including pyrolysis oil, carbon black, gas, light fuel fraction, and steel wires.

10.2. Key Contributions

One contribution of this study is the development of an evaluation framework for small-scale waste tyre pyrolysis system. Unlike most studies that consider the project's feasibility based on financial or technical factors, the current paper **uses simulation, economic analysis, and environmental considerations as key criteria**.

The second contribution of this research is to show the importance of applying discrete event simulation approach in pyrolysis plants' performance analysis. Most of the existing literature has focused **on chemical and thermodynamics modeling**. However, there is a dearth of research that explores other aspects of operation such as productivity, resource consumption, queue management, and bottlenecking.

Third, the contribution of this research is **the identification of the key factor behind the efficiency of the process** – the pyrolysis reactor, which controls product flow, financial benefit, and waste management capacity of the process.

Fourth, the research contributes to the body of knowledge by developing a preliminary **economic analysis model** that allows estimating capital investment cost, operational cost, revenues, payback period, ROI, and sensitivity analysis.

Finally, this study contributes to the field by providing practical implementation of **circular economy concepts**, where waste tyres become products of value.

10.3. Recommendations for Industry

Based on the analysis of results, the following recommendations are applicable in the industry.

First, there should be attention paid to improving the performance of the **reactor** as it appears to be the limiting factor. Reactor cycle time reduction, improving the processes of loading/unloading, or installing another reactor would help increase the capacity.

Second, there should be maintained an effective buffer/batching strategy. Due to the fact that the **reactor requires** a full batch to start operating, inappropriate batching will result in waiting, stacking, and reactor idle time.

Third, investments **should not be solely based on optimism regarding future profits**. A comprehensive market study has to be performed in order to establish **actual selling prices and demand levels** of pyrolysis oil, carbon black, steel wires, and light fuel fraction.

Fourth, special attention should be paid to **product quality management**. Economic value of pyrolysis products highly depends on their quality level, namely pyrolysis oil and carbon black, and further upgrading/treatment might be necessary.

Fifth, the issue of **environmental regulations** should be taken into account during project planning stage since the costs associated with emissions, gas treatment, safety equipment, permits, and waste handling should be estimated and incorporated.

Sixth, the **preventive maintenance technique** should be utilized in order to minimize downtime periods because the reactor is the source of all production.

Seventh, it should be maximally utilized that **pyrolysis gas** can be used as a heating resource.

10.4. Future Research Directions

Future research can extend this study in several directions.

First, further work can be done with data **on the operation of existing facilities** to enhance the accuracy of processing time, machine capacities, downtime, product yield, and economic analysis.

Second, further studies can focus on **creating a process model that considers such factors** as temperature, residence time, reactor type, catalyst usage, and composition of tyres as factors influencing the yield and quality of the final products.

Third, future investigations can be carried out to **examine the performance of discrete-event simulation combined with chemical processes simulation** tools like Aspen Plus.

Fourth, a life cycle assessment could be performed to **evaluate more precisely the influence of various factors on greenhouse gas emissions**, energy consumption, avoided emissions, benefits from materials recovery, and comparisons with alternative treatment technologies such as landfilling or incineration.

Fifth, further research may consider various configurations of pyrolysis systems involving **the use of two reactors**, increase in reactor capacity, change in batch sizes, adding extra shredders, and better buffering strategies.

Sixth, uncertainty analysis can be conducted based on the additional factors **like fluctuations in the price of products, equipment breakdowns and downtime, inflation, increasing labor costs, higher maintenance costs, and possible regulatory risks**.

Seventh, further investigations could be done into **possibilities of product upgrading** in order to raise their value and make them more marketable.

Finally, further research may produce a decision-making tool to choose the optimal configuration of the waste tyre pyrolysis plant based **on simulation results, economics, and environmental impact**.

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12. Appendices

12.1. Appendix A: Source of Project Data

The principal data on operation and economics in the context of the current investigation have been collected from the **Bio Fuel Company**. This organization has been chosen as a reference point for the development of small-scale waste tyre pyrolysis technology.

This organization supplied relevant data for the establishment of capacity, process flow diagram, equipment, assumptions on operations, and parameters concerning economic aspects of the investigated system **as shown in the figure below**. Such data included information on capacity for daily processing of tyres, average weight of one tyre, batch capacity of reactors, preprocessing of tyres, man-hours required for one batch, operating costs, assumptions on capital investments, and product outputs.

In order to conduct a preliminary analysis, all data gathered from the Bio Fuel Company have been supplemented by additional literature assumptions when necessary.



BioFuel

BioFuel Company Data Report

Production and Operational Data for the Small-Scale Waste Tyre Pyrolysis System

Prepared as an official company data reference.

Table 1: General Production Data

This table summarizes the main production characteristics of the simulated system.

Parameter	Value
Average Tire Weight	10 kg
Tires per Day	1000–1500 Tires
Daily Production	10–15 tons/day
Pieces per Tire	4 Pieces
Total Pieces per Day	4000–6000 Pieces

Table 2: Processing Times and Capacities

This table defines the processing times and capacity constraints of all system components.

Process	Processing Time	Unit	Capacity
Debeading	Random.Triangular(0.2, 0.3, 0.5)	min/tire	1
Cutting	Random.Triangular(0.15, 0.25, 0.4)	min/tire	1
Shredding	Random.Triangular(0.25, 0.4, 0.6)	min/piece	1
Buffer	—	—	Large
Reactor (Oven)	Random.Triangular(18, 20, 23)	hours/batch	10 tons



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General Manager
BioFuel Company

12.2. Appendix B: Production and Process Data

The production data used in the simulation model were based on the practical information provided by **Bio Fuel Company**. These data were used to define the material flow and production scale of the proposed plant.

Parameter	Value
Average tyre weight	10 kg
Daily tyre input	1000–1500 tyres/day
Daily production capacity	10–15 tons/day
Pieces per tyre after cutting	4 pieces
Total pieces per day	4000–6000 pieces/day
Reactor batch capacity	10 tons/batch
Required tyres per batch	1000 tyres
Required pieces per batch	4000 pieces
Reactor cycle time	18–23 hours/batch

These values were used to configure the discrete-event simulation model and to represent the interaction between continuous preprocessing stages and batch-based pyrolysis operation.

12.3. Appendix C: Simulation Input Parameters

The simulation model was developed using Simio software. The processing times and capacities were assigned based on the system structure and production assumptions obtained from Bio Fuel Company.

Process Stage	Processing Time	Unit	Capacity
Debeading	Random.Triangular(0.2, 0.3, 0.5)	min/tyre	1 tyre
Cutting	Random.Triangular(0.15, 0.25, 0.4)	min/tyre	1 tyre
Shredding	Random.Triangular(0.25, 0.4, 0.6)	min/piece	1 piece
Buffer	No processing time	Waiting/storage only	Large
Pyrolysis Reactor	Random.Triangular(18, 20, 23)	hours/batch	10 tons

The triangular distributions were selected to represent processing-time variability under practical operating conditions. These distributions allow the model to capture minimum, most likely, and maximum processing times for each operation.

12.4. Appendix D: Batching Logic Used in the Simulation Model

The batching logic was one of the most important elements in the simulation model because the pyrolysis reactor operates in batch mode.

The logic was developed as follows:

Item	Value
Average tyre weight	10 kg
Reactor capacity	10 tons
Number of tyres required per batch	1000 tyres
Number of pieces per tyre	4 pieces
Number of pieces required per batch	4000 pieces

Therefore:

1 tyre = 4 pieces

1000 tyres = 10 tons

4000 pieces = 1 reactor batch

In the simulation model, shredded tyre pieces accumulate in the buffer until 4000 pieces are available. Once this quantity is reached, the combiner forms one complete batch and sends it to the pyrolysis reactor.

This logic was necessary to represent the difference between continuous preprocessing operations and batch-based reactor operations.

12.5. Appendix E: Economic Data and Financial Assumptions

The economic data used in the feasibility analysis were mainly based on the practical information obtained from **Bio Fuel Company**, supported by market-based assumptions where needed.

Economic Parameter	Value
Estimated initial investment	700,000 JD
Monthly operating cost	7,000–10,000 JD/month
Daily operating cost	233–333 JD/day
Expected daily product profit	2,150–2,850 JD/day
Expected net daily profit	2,217 JD/day
Expected annual net profit	798,120 JD/year
Expected payback period	0.88 years
Expected ROI	114%

The economic feasibility analysis was developed to evaluate whether the project can generate sufficient financial return under different operating scenarios. Three scenarios were considered: conservative, expected, and optimistic.

12.6. Appendix F: Product Yield Assumptions

The output distribution of the pyrolysis process was based on the project data and typical product yield assumptions for waste tyre pyrolysis systems.

Product	Approximate Yield
Pyrolysis oil / diesel-like product	40%
Carbon black	30–40%
Gas	10%
Light fuel fraction	10%
Steel wires	10%

The gas fraction was not treated as a direct sales revenue stream in the financial model. Instead, it was assumed to be reused internally as a heat source for the reactor. This reduces external fuel consumption and improves the overall operating efficiency of the plant.

12.7. Appendix G: Environmental Assessment Assumptions

The environmental assessment was based on the principle that waste tyre pyrolysis can support sustainable waste management by converting end-of-life tyres into useful products.

The environmental evaluation considered the following aspects:

Environmental Aspect	Description
Waste reduction	Reduces the accumulation of waste tyres in landfills and stockpiles
Resource recovery	Converts tyres into oil, carbon black, steel wires, gas, and light fuel
Energy recovery	Reuses pyrolysis gas as a heat source for the reactor
Circular economy	Treats waste tyres as recoverable resources rather than waste
Avoided environmental burden	Reduces the need for uncontrolled burning or improper disposal

The environmental analysis was conceptual and did not include a complete life cycle assessment. Therefore, the results should be considered preliminary and should be supported by detailed emission measurements in future work.

12.8. Appendix H: Simulation Model Outputs

The simulation model was used to evaluate the operational performance of the proposed waste tyre pyrolysis system. The main outputs considered in the analysis were:

Output Indicator	Purpose
Throughput	Measures the quantity of material processed by the system
Reactor utilization	Evaluates how intensively the reactor is used
Machine utilization	Measures the use of preprocessing equipment
Queue size	Identifies congestion points
Waiting time	Measures delays before processing stages
Buffer level	Evaluates material accumulation before batching
Bottleneck location	Identifies the main system limitation
Completed batches	Measures the number of reactor batches processed

The simulation results showed that the preprocessing stages can supply material continuously, while the pyrolysis reactor represents the main bottleneck due to its long batch cycle time.

12.9. Appendix I: Python-Based Financial Sensitivity Analysis

A Python-based financial sensitivity model was used to evaluate the effect of changes in operating cost and product profit on project feasibility. The model calculated the following indicators:

Financial Indicator	Description
Net daily profit	Daily product profit minus daily operating cost
Monthly net profit	Net daily profit multiplied by operating days per month
Annual net profit	Monthly net profit multiplied by 12 months
Payback period	Initial investment divided by annual net profit
ROI	Annual net profit divided by initial investment

The sensitivity analysis tested different scenarios, including increases in operating costs and decreases in product profit. This helped identify the financial risk of the project and evaluate its feasibility under less favorable conditions.

12.10. Appendix J: Notes on Data Reliability and Study Boundaries

The data used in this study were mainly obtained from **Bio Fuel Company** and were used as the practical foundation for the simulation and feasibility analysis. However, the study remains a preliminary evaluation and should not be considered a final investment decision.

Several limitations should be considered:

1. Actual operating conditions may differ from the assumed values.
2. Product prices may change depending on market demand and product quality.
3. Reactor downtime can significantly affect profitability.
4. Environmental performance depends on emission control systems and regulatory compliance.
5. The model does not include detailed chemical kinetics.
6. The environmental assessment does not include a full life cycle assessment.
7. The financial results depend strongly on the assumed CAPEX, OPEX, and daily product profit.

Therefore, future implementation should include detailed engineering design, supplier quotations, pilot testing, product quality analysis, environmental permitting, and market verification.

12.11. Appendix K: Summary of Data Source

All primary project data used in the simulation model, economic feasibility analysis, and environmental assessment were derived from information provided by **Bio Fuel Company**. The company data served as the main practical reference for defining the production scale, process assumptions, equipment requirements, cost structure, and expected product outputs of the proposed waste tyre pyrolysis system.

Where direct data were unavailable, reasonable assumptions were made based on project context and supported by literature values. This approach allowed the study to develop a realistic preliminary evaluation of the technical, economic, and environmental feasibility of the proposed system.